Virtual Reality Hazard Identification Safety Training Simulator (SecureSite VR)

SWE/141/24B

GERMAN-MALAYSIAN INSTITUTE

JANUARY 2026

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This project report is submitted as a partial fulfilment of the requirement for the Diploma in Software Engineering

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DEDICATION

We would like to dedicate this final year project to our beloved parents for their endless love, encouragement, and unwavering support throughout our academic journey. Their sacrifices, prayers, and motivation have been our greatest source of strength and inspiration in completing this project successfully.

We would also like to express our heartfelt appreciation to our TTOs and friends for their guidance, cooperation, and continuous encouragement. Their valuable advice and support have greatly contributed to the improvement and completion of this project.
Lastly, we dedicate this project to everyone who continuously works towards creating positive change and contributing to a safer and better future for society and our country.

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In addition, we would like to thank our classmates from DSWE 7 and all our friends for their cooperation, encouragement, and support throughout this journey.

Special appreciation is also dedicated to our families for their endless prayers, understanding, and moral support, which motivated us to overcome every challenge during the development of this project.

Lastly, we would like to thank everyone who contributed directly or indirectly to the success of this project. This achievement would not have been possible without the collective effort and support from all parties involved.

ABSTRACT

The SecureSite VR is a virtual reality-based training system developed to improve workplace safety awareness among new industrial workers. The system is designed to provide an interactive, immersive, and realistic training experience by simulating a warehouse environment containing common workplace hazards such as slippery floors, falling objects, blocked emergency exits, and forklift movement areas.

The simulator allows users to explore the virtual environment, identify hazards, and receive immediate feedback along with performance evaluation through a scoring system. The system is developed with usability, accessibility, and user engagement in mind to ensure trainees can improve their hazard recognition skills effectively in a safe and controlled environment without exposure to actual danger.

In addition, the system includes features such as user registration, training instructions, real-time feedback, performance tracking, and result summaries to support the learning process. This report presents the overall design, development, and implementation of the SecureSite VR, highlighting the use of virtual reality technology and a structured software development approach in creating an effective workplace safety training solution.

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CHAPTER I

INTRODUCTION

1.1 PROJECT BACKGROUND

Workplace safety is an important aspect in industrial environments such as warehouses, factories, and manufacturing areas. New industrial workers are often exposed to various workplace hazards, including slippery floors, falling objects, blocked emergency exits, forklift movement, and electrical risks. Due to lack of experience and limited hazard awareness, new workers are more likely to be involved in workplace accidents compared to experienced employees.

Most companies still rely on traditional safety training methods such as lectures, printed manuals, and videos. Although these methods provide basic knowledge, they are often less engaging and do not offer practical experience in identifying real workplace hazards. As a result, workers may find it difficult to apply safety knowledge in actual working conditions.

With the advancement of Virtual Reality (VR) technology, safety training can be improved through immersive and interactive learning. VR allows users to experience realistic workplace environments in a safe virtual setting without exposure to actual danger. Users can practice identifying hazards, receive immediate feedback, and improve their safety awareness through repeated training.

Therefore, this project proposes the development of a SecureSite VR for new industrial workers. The system aims to provide a more effective and engaging training method that improves hazard recognition skills and supports safer workplace practices.

1.2 PROBLEM STATEMENT

- Traditional workplace safety training methods such as lectures, manuals, and videos, are passive and less interactive.

- New industrial workers often lack experience and hazard awareness, increasing the risk of workplace accidents.

- Conventional training methods do not provide realistic practical exposure to workplace hazards.

- Workers may have difficulty identifying hazards effectively in actual working environments.

- Traditional safety training can lead to low engagement and poor knowledge retention among trainees.

- Many organizations still do not utilize Virtual Reality (VR) technology in safety training.

- There is a need for a more immersive, interactive, and effective safety training solution for hazard identification.

- Therefore, a SecureSite VR is proposed to improve hazard awareness and workplace safety training for new industrial workers.

1.3 PROJECT OBJECTIVES

The following objectives aim to address the key shortcomings, as outlined in the problem statement:

1. To develop a VR-based simulation system that allows users to identify hazards in an warehouse environment.

2. To display interactive hazard detection and scoring mechanisms for training evaluation.

3. To improve safety awareness and user engagement by allowing users to identify hazards and receive feedback in a VR simulation.

1.4 PROJECT SCOPE

This project focuses on the development of a Virtual Reality (VR) Hazard Identification Safety Simulator designed for new industrial workers undergoing safety training. The system simulates a warehouse environment where users can explore and identify common workplace hazards.

The scope of the project includes the creation of a single virtual warehouse scene with selected hazards such as slippery floors, falling objects, blocked emergency exits, and forklift movement areas. Users will be able to navigate the environment, interact with objects, and identify hazards through a simple selection mechanism. The system will provide immediate feedback and a scoring system to evaluate user performance.

In addition, the project includes basic user interface features such as a main menu, instructions page, and result summary screen. The simulator will be developed using Unity and supports both VR mode and desktop mode for accessibility.

However, the project does not include advanced features such as multiplayer functionality, real-time online database integration due to time and resource constraints. The system is intended for educational and training demonstration purposes only.

CHAPTER II

LITERATURE REVIEW

2.1 FEASIBILITY STUDIES

A feasibility study is conducted to determine whether the proposed project is practical and achievable. For the SecureSite VR, the feasibility study was carried out using both primary and secondary sources. The purpose of this study is to understand the needs of warehouse workers, evaluate the effectiveness of VR-based safety training, and determine whether the system is suitable and practical to be developed and implemented.

2.2 PRIMARY SOURCES

Primary sources refer to original data collected firsthand for a specific research purpose. These data are raw and have not been previously analyzed or interpreted. In this project, a survey technique was conducted to collect relevant information for the development of the system by distributing a questionnaire to 41 respondents.

2.2.1 Questionnaire

A questionnaire is a research instrument consisting of a series of questions designed to collect information from respondents regarding their experiences, opinions, and awareness levels. In this project, both close-ended and open-ended questions were applied to obtain quantitative and qualitative data related to workplace safety training.

The questionnaire was distributed to warehouse workers to identify common safety challenges, evaluate their understanding of workplace hazards, and measure their interest in using the SecureSite VR. The responses collected were used to analyse user requirements, safety awareness, and the overall acceptance and effectiveness of the proposed VR training system.

Below is the analysis of the data gathered from the questionnaires we distributed:

2.2.1.1 What is your role?

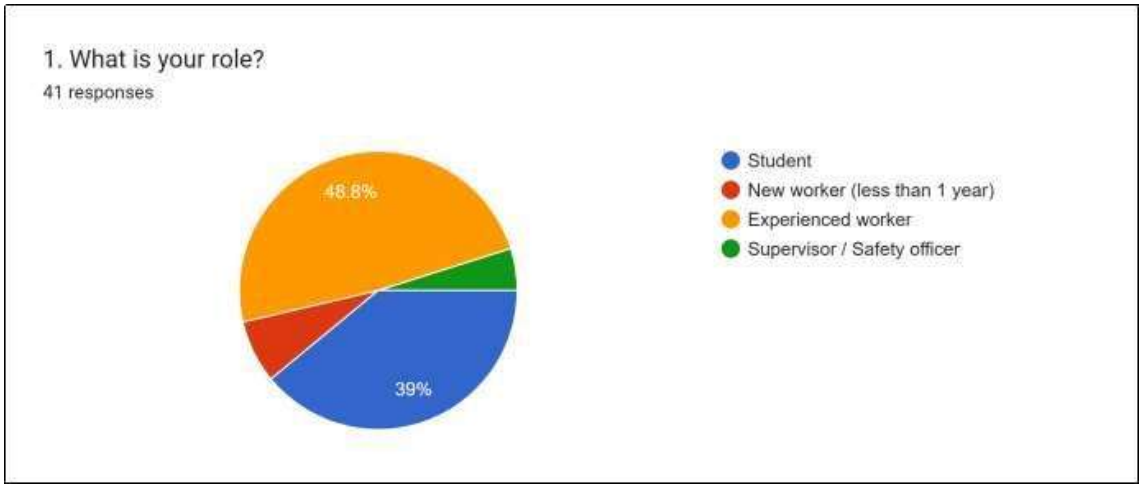


Figure 2.2.1.1: Questionnaire - Question 1

The pie chart above illustrates the respondents' current employment status. The majority of respondents, accounting for 48.8%, are experienced workers, followed by 39% students, while 7.3% are new workers, along with a smaller proportion of supervisors or safety officers.

Although experienced workers form the largest group, this study focuses on new employees as the target users. The presence of students and new workers in the data highlights the relevance of developing a system that supports individuals who are new to the workforce. Additionally, experienced workers and supervisors can provide valuable

insights into workplace practices, which can help improve the system's effectiveness for new employees.

2.2.1.2 Have you worked in a warehouse environment?

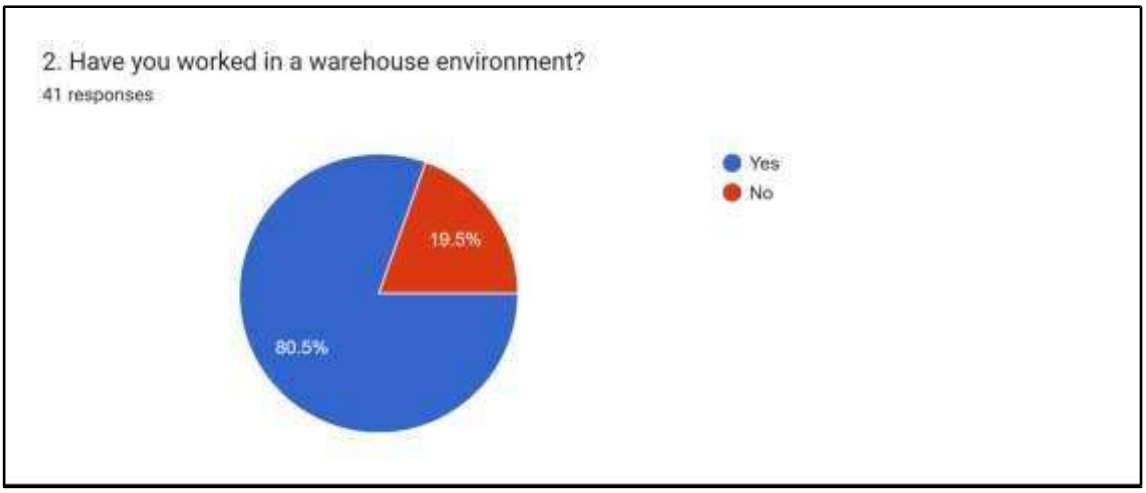


Figure 2.2.1.2: Questionnaire - Question 2

The pie chart shows the distribution of respondents based on their experience working in a warehouse environment. Based on the chart, a total of 41 responses were collected. The majority of respondents, 80.5%, answered "Yes," indicating that they have worked in a warehouse environment. Meanwhile, 19.5% answered "No," showing that a smaller portion of respondents do not have warehouse experience.

The large percentage of respondents with warehouse experience indicates that most participants are familiar with common workplace hazards and safety procedures in warehouse operations. Their practical knowledge makes the survey results more reliable and relevant for this project.

This finding supports the decision to use a warehouse environment as the main simulation setting for the VR system. Since most respondents have direct experience in this type of workplace, the project can better reflect real industry conditions and provide more effective hazard identification training for new industrial workers.

2.2.1.3 What type of warehouse industrial environment have you worked in?

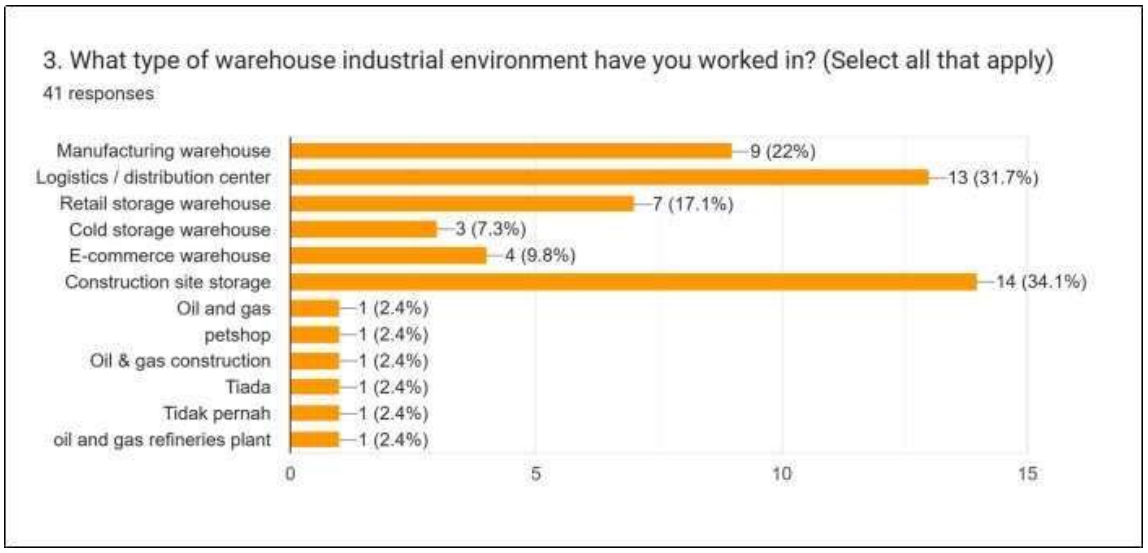


Figure 2.2.1.3: Questionnaire - Question 3

The bar chart above shows the types of warehouse or industrial environments in which respondents have worked. The most common environment is construction site storage at 34.1%, followed by logistics or distribution warehouses at 31.7%. Manufacturing warehouses account for 22% of respondents, while retail storage warehouses represent 17.5%. In addition, 9.8% of respondents have experience in e-commerce warehouses, and 7.3% have worked in cold storage facilities. A small proportion of respondents, 2.4%, reported experience in oil and gas-related environments, including refineries and construction projects.

Overall, the data indicates that respondents have diverse experience across various warehouse and industrial settings, with a strong concentration in logistics, distribution, and construction-related storage environments. This suggests that the findings are relevant to users working in different operational contexts within the warehouse and industrial sector.

2.2.1.4 How long is your experience?

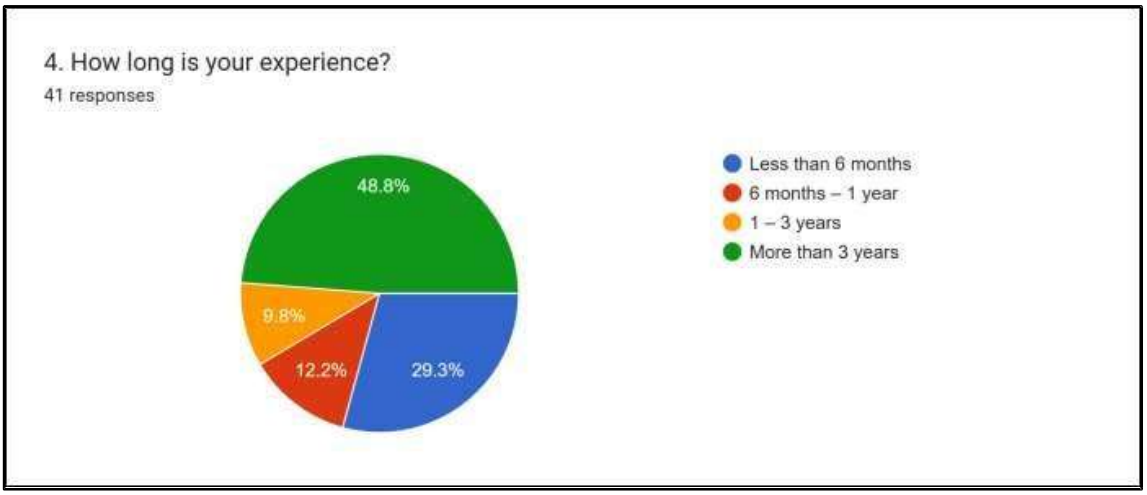


Figure 2.2.1.4: Questionnaire - Question 4

The pie chart above illustrates the respondents' length of work experience. The largest group of respondents has more than 3 years of experience, accounting for 48.8%. This is followed by those with less than 6 months of experience, at 29.3%. Respondents with 6 months to 1 year of experience represent 12.2%, while those with 1 to 3 years of experience make up 9.8%.

Overall, the data indicate that respondents include both highly experienced individuals and newcomers to the workforce, with a significant proportion having long-term work experience.

2.2.1.5 Have you received safety training before starting work?

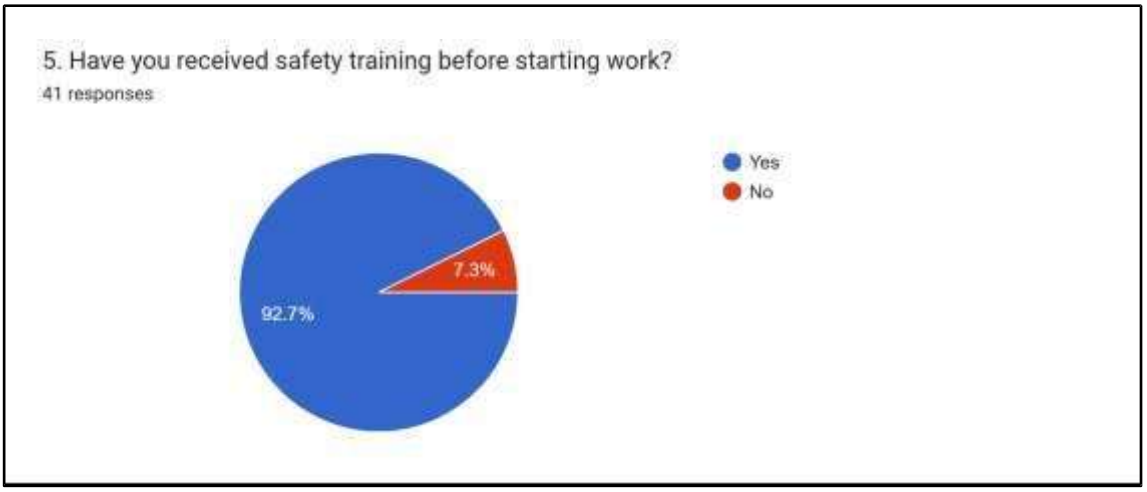


Figure 2.2.1.5: Questionnaire - Question 5

The pie chart shows the distribution of respondents based on whether they have received safety training before starting work. This question was included to understand how common safety training is among workers before entering a warehouse or industrial environment. Based on the chart, a total of 41 responses were collected. The majority of respondents, 92.7%, answered “Yes,” indicating that they had received safety training before starting work. Meanwhile, only 7.3% answered “No,” showing that a small number of respondents started work without formal safety training.

However, although most respondents received training, traditional methods such as lectures, manuals, and videos may not always provide enough practical experience in identifying real hazards. This supports the need for a VR-based safety training system that offers a more interactive and realistic learning experience. The proposed SecureSite VR can help improve hazard awareness by allowing users to practice in a safe virtual environment before entering the actual workplace.

2.2.1.6 What type of safety training did you receive?

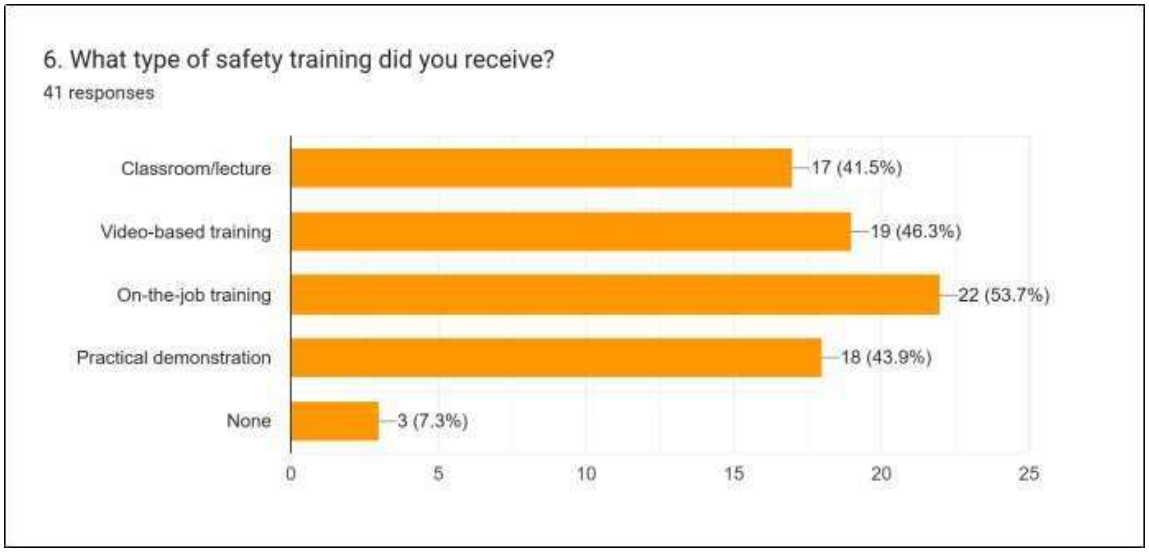


Figure 2.2.1.6: Questionnaire - Question 6

The bar chart above shows the types of safety training received by respondents. The most common type of training is on-the-job training, accounting for 53.7% of respondents. This is followed by video-based training at 46.3%, practical demonstration at 43.9%, and classroom or lecture-based training at 41.5%. A small proportion of respondents, 7.3%, reported that they did not receive any safety training.

Overall, the data indicates that most respondents have received some form of safety training, with on-the-job training being the most prevalent method. This suggests that practical, work-based learning is commonly used in industrial and warehouse environments.

2.2.1.7 How would you rate the effectiveness of your safety training?



Figure 2.2.1.7: Questionnaire – Question 7

The bar chart shows respondents' ratings of the effectiveness of their safety training using a scale from 1 to 5, where 1 represents very poor and 5 represents excellent. This question was included to evaluate how effective current safety training methods are in helping workers understand workplace hazards and safety procedures.

Based on the chart, a total of 41 responses were collected. The highest number of respondents, 53.7% (22 respondents), rated the effectiveness of their safety training as 5, indicating excellent satisfaction. This is followed by 29.3% (12 respondents) who rated it as 4, showing that many respondents found the training effective. Meanwhile, 9.8% (4 respondents) selected 3, while only 4.9% (2 respondents) rated it as 1 and 2.4% (1 respondent) rated it as 2.

The results show that most respondents consider their current safety training to be effective. This suggests that safety training is already valued and implemented in many workplaces. However, even with positive ratings, traditional training methods may still have limitations in providing practical and immersive experiences for hazard identification.

2.2.1.8 What problems did you face during safety training?



Figure 2.2.1.8: Questionnaire - Question 8

The bar chart above shows the problems faced by respondents during safety training. Most respondents, 56.1%, reported that they did not face any problems during their training. However, some respondents identified several issues, including boring or less engaging training and lacks practical experience 19.5%. In addition, 17.1% of respondents found it hard to imagine real situations, while another 17.1% felt that the training was too theoretical. A small number of respondents, 7.3%, said that the training was difficult to understand.

Overall, the data show that although most respondents did not face problems, some still experienced issues related to engagement and practical learning. This suggests that more interactive and hands-on training methods could improve training effectiveness.

2.2.1.9 Have you ever encountered or witnessed workplace hazards?

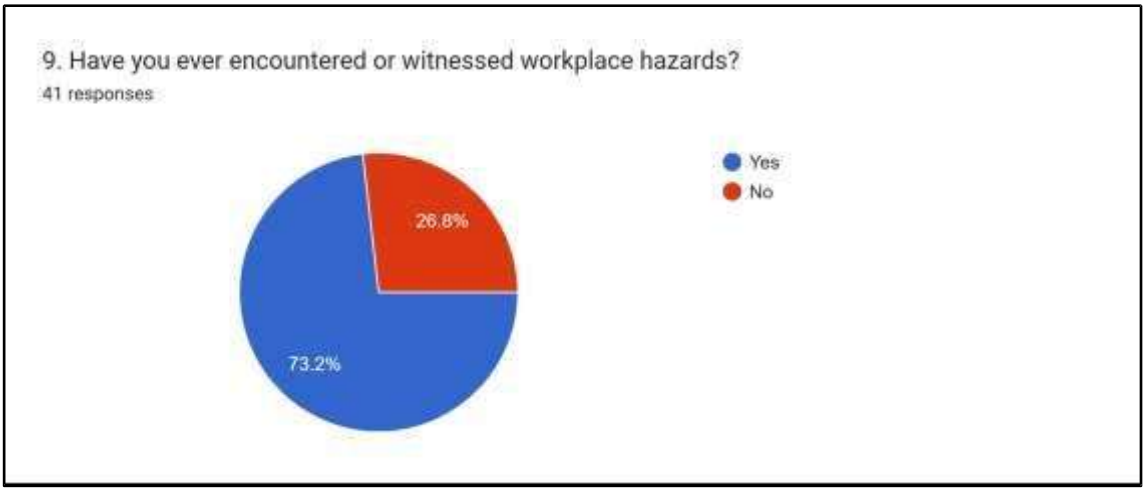


Figure 2.2.1.9: Questionnaire - Question 9

The pie chart above shows regarding personal encounters with workplace hazards reveal a significant prevalence of safety risks in industrial environments. Out of 41 respondents, a substantial 73.2% reported that they have personally encountered or witnessed workplace hazards, while only 26.8% indicated otherwise. This high percentage of exposure underscores the reality that hazards are not merely theoretical possibilities but frequent occurrences in the field.

The findings suggest that while traditional safety training exists, a vast majority of workers are still coming into direct contact with dangerous conditions. By implementing a VR simulation that focuses on the most common hazards identified in this study such as forklift traffic and blocked emergency exits.This project addresses a documented high frequency problem, providing new workers with a safe, immersive environment to develop the recognition skills needed to navigate these common industrial risks.

2.2.1.10 What types of hazards have you seen?

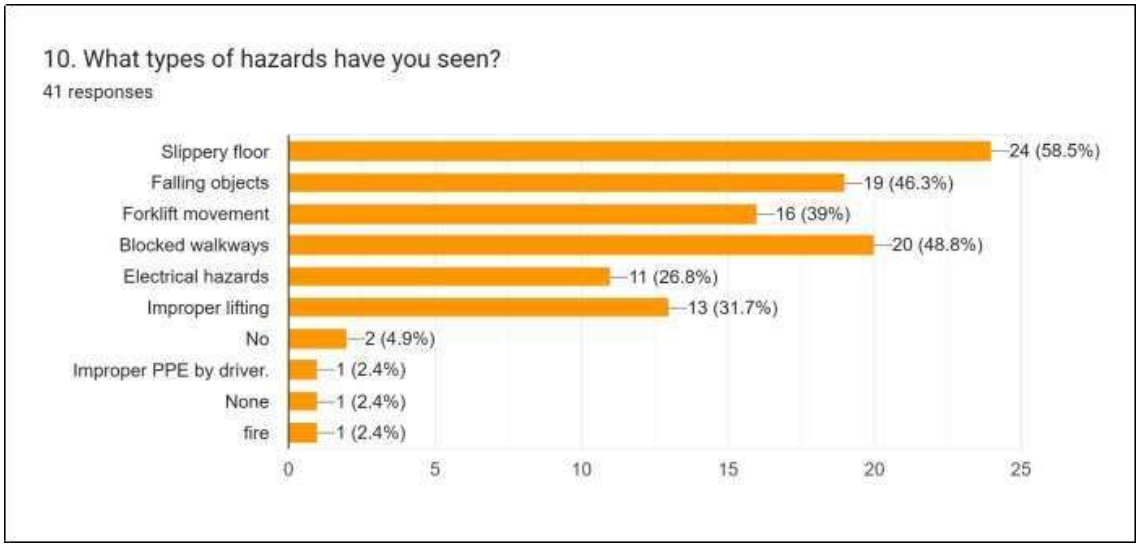


Figure 2.2.1.10: Questionnaire - Question 10

The bar chart above shows the types of hazards experienced by respondents. The most common hazard is slippery floors, reported by 58.5% of respondents. This is followed by blocked walkways at 48.8% and falling objects at 46.3%. Forklift movement was reported by 39% of respondents, while improper lifting accounted for 31.7%. Electrical hazards were experienced by 26.8% of respondents. A small proportion of respondents, 2.4%, reported other hazards such as improper PPE use by drivers and fire incidents, while some respondents indicated none.

Overall, the data shows that slippery floors, blocked walkways, and falling objects are the most common hazards in the workplace. This highlights the importance of strong safety measures to reduce these risks.

2.2.1.11 How confident are you in identifying workplace hazards?



Figure 2.2.1.11: Questionnaire - Question 11

The bar chart above shows respondents confidence in identifying workplace hazards using a Likert scale. A small proportion of respondents selected level 1 (4.9%) and level 2 (2.4%), indicating low confidence. Meanwhile, 22% of respondents selected level 3, showing a moderate level of confidence. A larger proportion selected level 4 (36.6%) and level 5 (34.1%), indicating high confidence in identifying workplace hazards.

Overall, the data suggests that most respondents are confident in identifying workplace hazards, with the majority rating themselves at a high level of confidence.

2.2.1.12 Have you ever used Virtual Reality (VR) before?

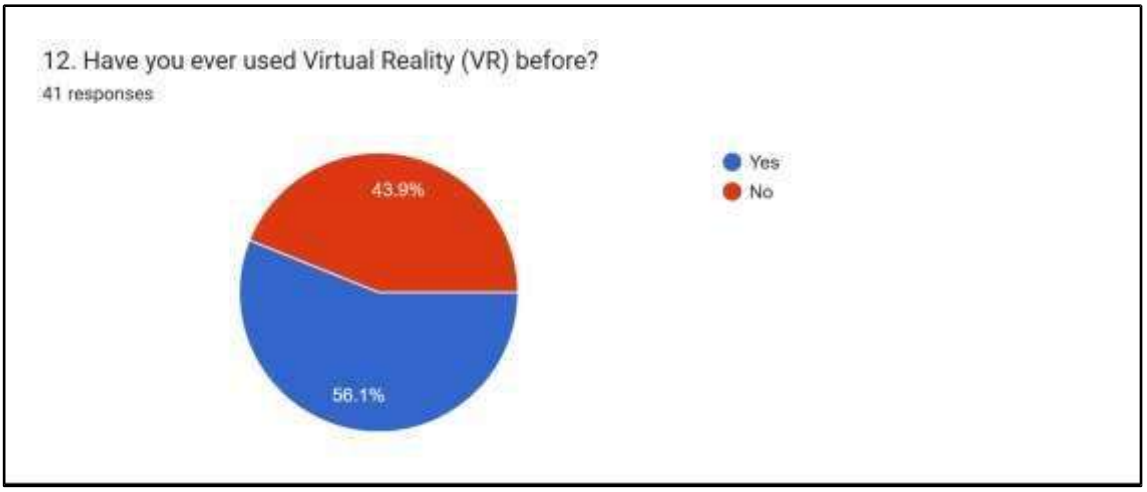


Figure 2.2.1.12: Questionnaire - Question 12

The pie chart above displays the respondents prior experience with Virtual Reality (VR) technology. The data shows that 56.1% of respondents have used VR before, while 43.9% of respondents have never utilized the technology.

Overall, the results suggest a relatively balanced level of technical familiarity among the target users. While more than half are already comfortable with the medium, the significant percentage of first-time users indicates that the simulator's interface and instructions must be designed with high accessibility and ease of use in mind.

2.2.1.13 Do you think VR can improve safety training?

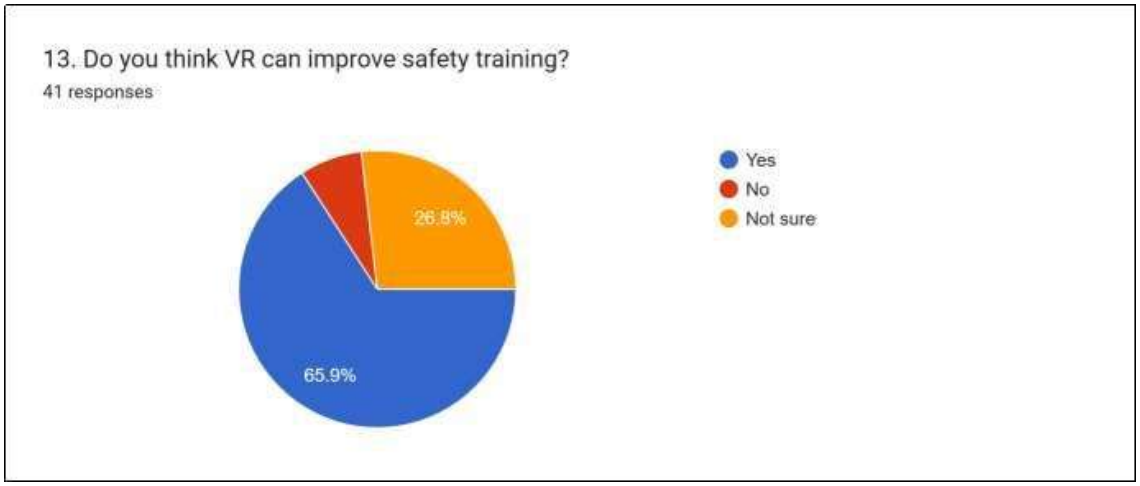


Figure 2.2.1.13: Questionnaire - Question 13

The pie chart above shows respondents' opinions on whether VR can improve safety training. The majority of respondents, 65.9%, believe that VR can improve safety training. Meanwhile, 26.8% of respondents are not sure, and a small proportion of 7.3% do not agree that VR can improve safety training.

Overall, the data indicates that most respondents have a positive view of using VR in safety training, suggesting strong potential acceptance of VR-based training systems.

2.2.1.14 What features would you like in a VR safety training system?

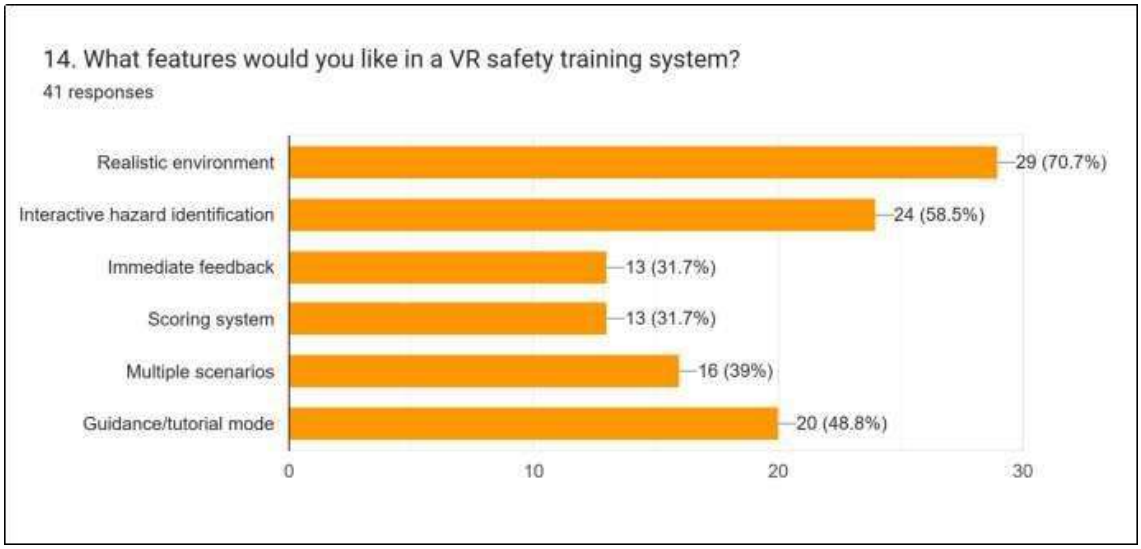


Figure 2.2.1.14: Questionnaire - Question 14

The bar chart above shows the features that respondents would like in a VR safety training system. The most preferred feature is a realistic environment, selected by 70.7% of respondents. This is followed by interactive hazard identification at 58.5% and guidance or tutorial mode at 48.8%. Multiple scenarios were chosen by 39% of respondents, while both immediate feedback and a scoring system were selected by 31.7% of respondents.

Overall, the data indicates that respondents prefer VR training features that are realistic and interactive. This suggests that a VR safety training system should focus on creating real-life simulations and engaging learning experiences to improve effectiveness.

2.2.1.15 What type of scenario would you like to experience in VR training?



Figure 2.2.1.15: Questionnaire - Question 15

The bar chart above shows the types of scenarios that respondents would like to experience in VR safety training. The most preferred scenario is hazard identification in a warehouse environment, selected by 80.5% of respondents. This is followed by proper lifting and manual handling, and emergency response (first aid or accident situations), both at 65.9%. Fire emergency and evacuation scenarios were chosen by 58.5% of respondents. In addition, forklift safety and awareness, as well as chemical hazardous material handling, were selected by 53.7% of respondents. Electrical hazard scenarios were chosen by 48.8% of respondents.

Overall, the data indicates that respondents prefer realistic and practical VR training scenarios related to common workplace hazards and emergency situations. This suggests that VR training should focus on real-world warehouse safety challenges to improve learning effectiveness.

2.2.1.16 Any additional comments or suggestions?



Figure 2.2.1.16: Questionnaire - Question 16

The question above aimed to gather respondents' additional comments and suggestions regarding safety training and the proposed system. Overall, most respondents gave positive feedback, stating that current safety training is informative and well-structured. They also mentioned that safety practices are generally good and should be continuously maintained.

However, several suggestions were made for improvement. Respondents recommended adding more real-life warehouse scenarios and practical training to improve understanding and hazard identification. They also highlighted the need for regular safety checks, continuous updates, and better communication between management and workers to strengthen workplace safety.

In addition, some respondents suggested that VR-based training could help improve learning by creating realistic scenarios and enhancing practical skills, especially in hazard identification and safety equipment usage. Overall, the feedback shows that safety training can be further improved through more interactive and practical approaches.

2.2.1.17 What improvements would you suggest for current safety training?

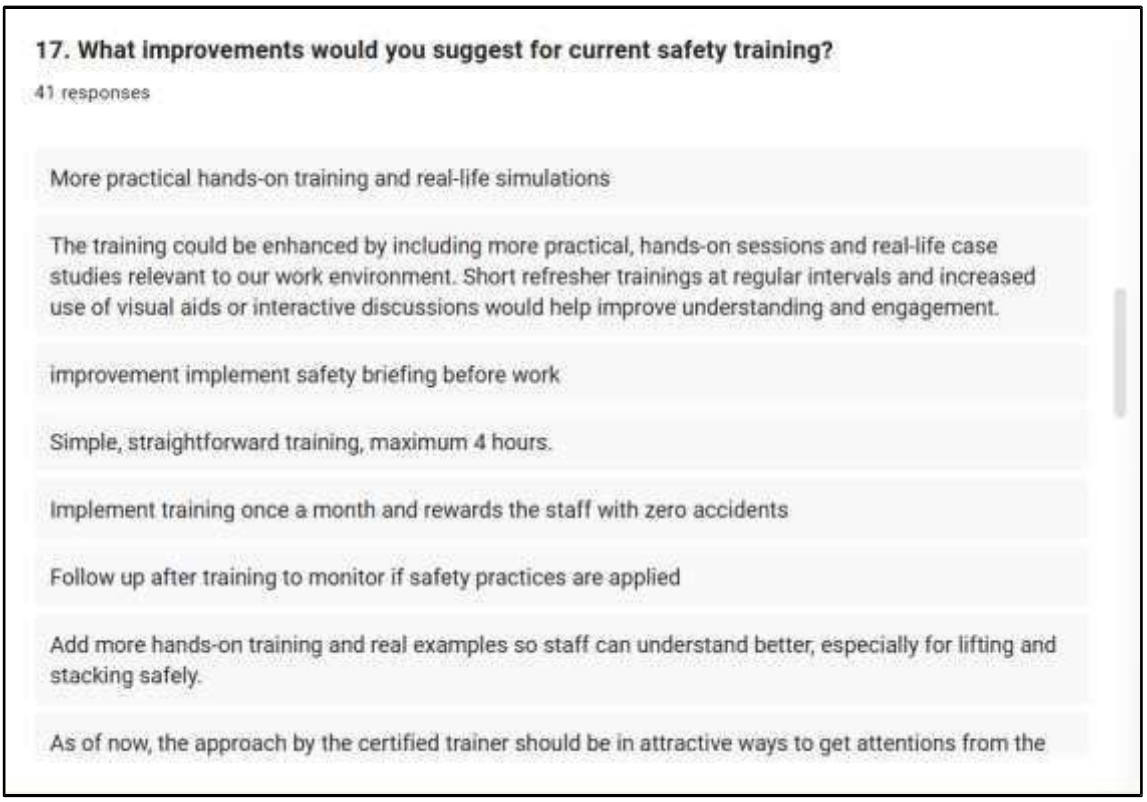


Figure 2.2.1.17: Questionnaire - Question 17

The question above aimed to gather respondents' suggestions for improving current safety training. Overall, respondents suggested that safety training should be conducted more frequently and on a regular schedule to continuously improve safety awareness among employees.

Many respondents recommended adding more practical, hands-on training and real-life simulations. They also suggested using real workplace examples, case studies, visual aids, and interactive sessions to make training more engaging and easier to understand.

In addition, respondents highlighted the importance of on-the-job training, real demonstrations, and routine hazard identification activities. Overall, the feedback shows that safety training should be more practical, interactive, and closely related to real working situations.

2.3 SECONDARY SOURCES

2.3.1 Online Articles

1. Traditional Safety Training Methods

Workplace safety training is commonly conducted through classroom lectures, printed manuals, and instructional videos. Although these methods provide theoretical knowledge, they are often passive and lack practical exposure to real industrial hazards. According to Xiaotong Guo. (2024), traditional training methods may not effectively improve hazard identification performance because learners do not experience realistic environments. As a result, workers may struggle to apply safety knowledge in actual workplace situations.



Figure 2.3.1.1: Secondary Sources – Article 1

2. Virtual Reality in Safety Training

Virtual Reality (VR) technology has emerged as an innovative solution to improve safety training effectiveness. VR allows users to experience immersive and interactive simulations of real workplace environments without facing actual risks. Conducted a systematic review and found that VR-based safety training significantly improves safety behaviour and skill acquisition compared to traditional approaches. Similarly, Md Shamsudin and Abdul Majid (2019) reported that trainees using VR simulation performed better in identifying construction hazards than those trained using conventional site visits. These findings show that VR enhances engagement, knowledge retention, and hazard recognition performance.

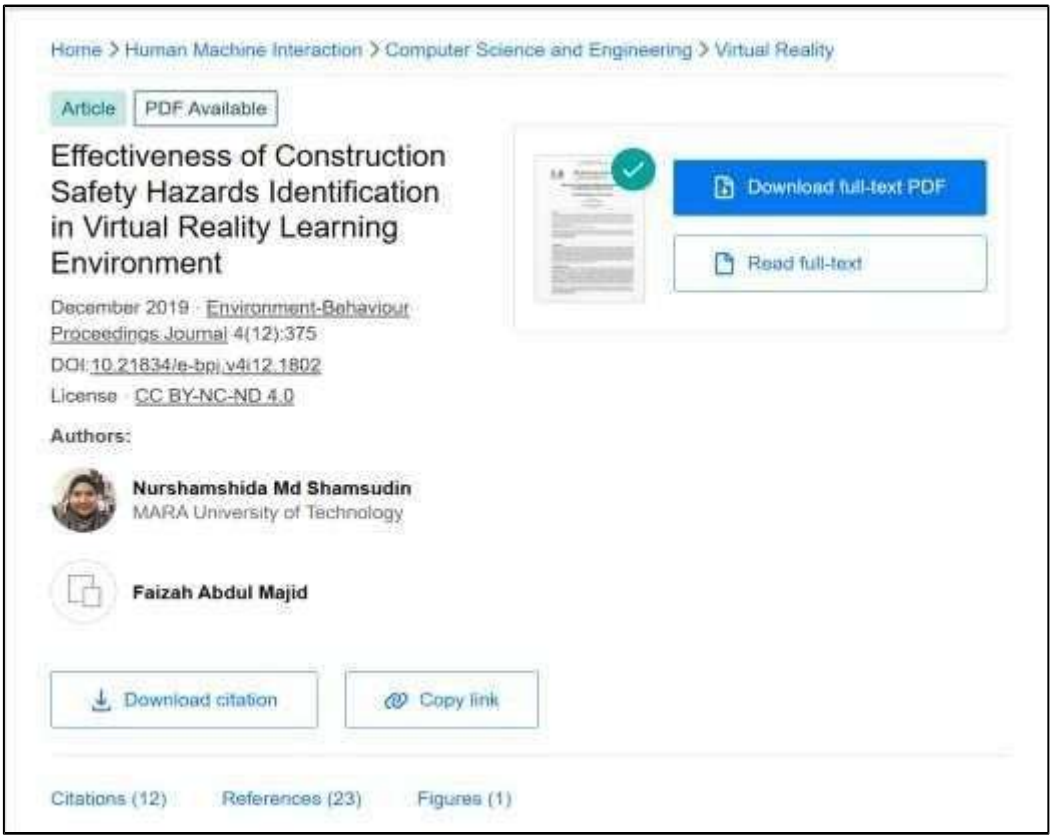


Figure 2.3.1.2: Secondary Sources - Article 2

3. Research Gap

Although previous studies by Bekir Enez Ozel. (2025) and Mehmet Koray Pekeriqli. (2025) have demonstrated the effectiveness of VR in safety training, many industrial workplaces still rely on traditional training methods. There is limited implementation of VR-based hazard identification systems specifically designed for interactive industrial training environments. Therefore, this project aims to develop SecureSite VR to provide a more engaging, realistic, and effective training solution for industrial workers.



Figure 2.3.1.3: Secondary Sources - Article 3

2.4 PRODUCT COMPARISON

This section compares existing Virtual Reality (VR) safety training solutions to evaluate their features, hardware requirements, strengths, weaknesses, and cost implications. The comparison helps determine the feasibility and competitive positioning of the proposed SecureSite VR project.

Comparison Criteria:

The following criteria were used for comparison:

i. Target Use Cases - Industries or environments the solution is designed for.
ii. Hardware Required - Type of VR equipment needed for operation.
iii. Key Features - Core functionalities such as scenario branching and analytics.
iv. Strengths - Competitive advantages of the product.
v. Weaknesses - Limitations or potential drawbacks.
vi. Price Range - Estimated relative cost level.

2.4.1 Product Comparison Tables

Product	Target Use Cases	Hardware Required	Strengths	Weaknesses	Price Level
Strivr	Industrial & enterprise safety training	High-end VR systems	Strong analytics & enterprise support	High setup and licensing cost	RM79,340 – RM396,700.00+
ForgeFX Simulations	Construction & hazard training	PC VR & Standalone VR	Highly customizable scenarios	Longer development timeline	RM39,670 – RM198,350
Our Project SecureSite VR	Tailored hazard-specific training	Configurable (Standalone or PC VR)	Full customization & domain specialization	Higher initial development effort	RM3,393 – RM 3,967 (estimated)

Table 2.4.1.1: Product Comparison Tables

2.4.2 Features Comparison Tables

Feature	*Strivr*	*ForgeFX Simulations*	*Our SecureSite VR*
Hazard Scenarios	✅ *Fire and explosion (Broad Industrial)*	✅ *(Construction/Hazard)*	✅ *(Warehouse-Specific)*
Interactivity Level	✅ *(High)*	✅ *(High)*	✅ *(High / Focused)*
Realism / Immersion	✅ *(High)*	✅ *(High)*	✅ *(Moderate to High)*
Performance Tracking	✅ *(Advanced Analytics)*	❌ *(Custom/Varies)*	✅ *(Scoring/Results)*
Hardware Accessibility	❌ *(Requires High-End)*	✅ *(PC & Standalone)*	✅ *(PC & Meta Quest 3)*
Customization	❌ *(Limited by License)*	✅ *(Highly Customizable)*	✅ *(Fully Tailored)*
User Feedback System	✅ *(Strong Analytics)*	✅ *(Simulation Feedback)*	✅ *(Immediate Feedback)*

Table 2.4.2.1: Features Comparison Tables

Based on the comparison tables, commercial enterprise solutions offer strong analytics and advanced features but are expensive. Standalone VR solutions are more affordable and easier to use. Customization is important for hazard-specific training, while analytics and performance tracking help improve training effectiveness. Overall, a mid-range custom standalone VR solution provides the best balance between cost, features, and feasibility.

Key Findings:

i. Commercial enterprise solutions provide robust analytics but involve high costs. Standalone VR solutions offer better affordability and easier deployment.
ii. Customization is a major advantage for niche hazard-specific training. Analytics and performance tracking can be a key differentiator.
iii. A mid-range custom solution optimized for standalone VR offers balanced feasibility.

2.4.3 Conclusion

Based on the comparison, developing a custom SecureSite VR is feasible and competitive if it focuses on cost-effective hardware deployment, hazard-specific realism, and integrated performance analytics. While commercial solutions dominate enterprise-level markets, there is strong potential for a specialized and scalable training simulator tailored to specific industry needs.

CHAPTER III

PROJECT METHODOLOGY

3.1 PRINCIPAL OF OPERATION

The principle of operation for the SecureSite VR explains how the system functions from the beginning of the training session until the final result is displayed. The system is designed to provide an interactive safety training experience for new industrial workers through a virtual warehouse environment.

First, the user launches the simulator and accesses the main menu. The user selects the “Start Training” option to begin the session. Before entering the simulation, the system displays training instructions, including the objective of identifying workplace hazards and the controls used for navigation and interaction.

Next, the user enters the virtual warehouse environment where common workplace hazards are placed, such as slippery floors, blocked emergency exits, exposed wires, and falling objects. The user can move around the environment and observe surrounding conditions.

When the user identifies a suspected hazard, they select the object using the interaction system. The system checks whether the selected object is an actual hazard. If the selection is correct, positive feedback is displayed and score points are added. If the selection is incorrect, the system provides warning feedback and may deduct points.

The system continues tracking user actions, including correct hazard identification, incorrect attempts, and time taken to complete the training. Once all hazards are found or the time limit is reached, the system displays a result summary showing total score, hazards identified, and hazards missed.

Finally, the user can choose to replay the training session for improvement or exit the system. This operation flow helps users improve hazard awareness through repeated practice in a safe and controlled virtual environment.

3.1.1 SDLC Process

This project uses the Agile Model as the Software Development Life Cycle (SDLC) approach. The Agile model is suitable because the project has clear objectives, fixed requirements, and a structured development flow. Each phase is completed step by step before moving to the next phase, making the project easier to manage and monitor.

Phases of Agile ll Model

1. Requirement Analysis

In this phase, information is collected through surveys and literature review to understand current workplace safety training problems. The project scope, user requirements, and hazard scenarios are identified.

2. System Design

This phase involves preparing the system architecture, use case diagram, flowchart, interface design, and hazard interaction logic. The warehouse environment and training process are planned before development begins.

3. Implementation

The VR simulator is developed using Unity and C# programming language. The virtual warehouse environment, hazard objects, scoring system, and user interface are created during this phase.

4. Testing

System testing that apply Unit Testing, Integration Testing, User Acceptance Testing (UAT) and Black-Box Testing method are conducted to ensure all functions work correctly. Hazard detection, scoring accuracy, feedback display, and user interaction are tested to reduce errors and improve system reliability.

5. Deployment and Documentation

The completed VR safety training simulator is deployed on a standalone VR platform such as Meta Quest using the Unity build application. After deployment, the system is demonstrated and tested by users in the actual training environment. Complete documentation, screenshots, diagrams, and reports are also finalized for submission.

6. Maintenance

After deployment, the system is maintained to fix bugs, improve performance, and update hazard scenarios when needed. User feedback is also collected to enhance the VR training experience and ensure the system remains effective and reliable.

3.1.2 Use Case Diagram

The Use Case Diagram shows how the main user interacts with the system. The main actor for this project is the Trainee (New Warehouse Worker) who uses the VR simulator for hazard identification training.

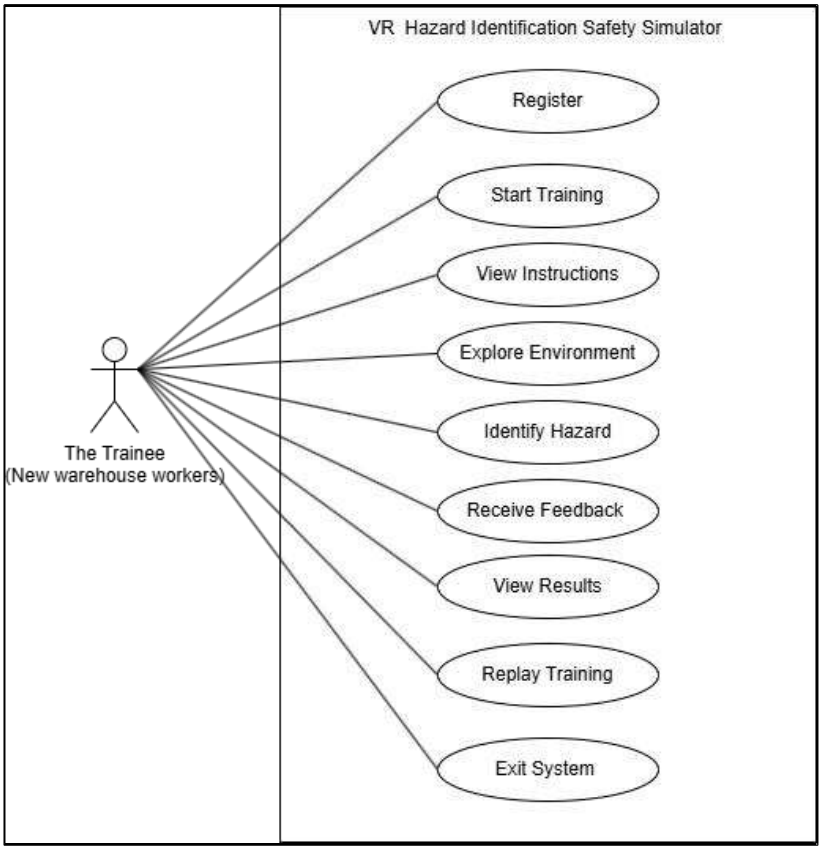


Figure 3.1.2: SecureSite VR - Use Case Diagram

The trainee starts the training session by accessing the simulator and viewing the instructions. The user then enters the virtual warehouse environment and explores the surroundings to identify hazards. When a hazard is selected, the system provides immediate feedback and updates the score. After all hazards are found or time ends, the system displays the result summary. The user can choose to replay the training or exit the system.

3.1.3. Use Case Specification



Figure 3.1.3.1.1: Use Case – Register

3.1.3.1 Use Case 1: Register

Use Case ID	UC-01
Use Case Name	Register
Description	Allows a new trainee to register before accessing the SecureSite VR.
Actor	Trainee (New warehouse worker)
Pre-Conditions	I. User has access to the VR simulator system II. System is running properly
Basic Flow	1. Trainee opens the VR safety simulator 2. Selects “Register” 3. Enters required details (name, ID, password, etc.) 4. Submits registration form 5. System validates input 6. System creates trainee account 7. Confirmation message is displayed

Alterantive Flow	**A1:** Any required field is empty → system displays an error message requesting complete information **A2:** Trainee ID already exists → system prompts user to use another ID
Post-Condition	Trainee account is successfully created and ready to access the SecureSite VR.

Table 3.1.3.1: Use Case Specification – Register



Figure 3.1.3.1.2: Use Case - Start Training

3.1.3.2 Use Case 2: Start Training

Use Case ID	UC-02
Use Case Name	Start Training
Description	Allows the trainee to begin SecureSite VR training session.
Actor	Trainee (New warehouse worker)
Pre-Conditions	Trainee has registered or logged into the system VR simulator is funtioning properly
Basic Flow	1. Trainee opens the main menu 2. Selects “Start Training” 3. System loads the virtual warehouse environment 4. Training session begins 5. Trainee can move and interact within the environment
Alternative Flow	**A1:** System fails to load training environment → system displays an error message and prompts trainee to restart

	A2: VR device not connected → system requests trainee to reconnect the device
Post-Condition	The VR training session starts successfully and the trainee enters the virtual warehouse environment.

Table 3.1.3. 2: Use Case Specification – Start Training

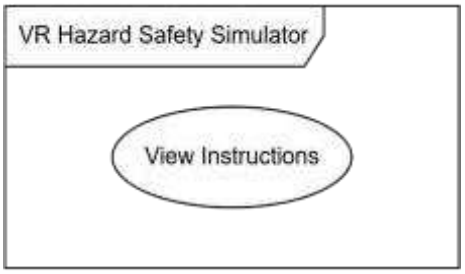


Figure 3.1.3.1.3: Use Case - View Instructions

3.1.3.3 Use Case 3: View Instructions

Use Case ID	UC-03
Use Case Name	View Instructions
Description	Allows the trainee to read and understanding the training guidelines before starting the VR simulation.
Actor	Trainee (New warehouse worker)
Pre-Conditions	I. Trainee has accessed the VR safety simulator II. Instruction module is available
Basic Flow	1. Trainee selects “View Instructions” 2. System displays training guidelines and controls 3. Trainee reads the instructions 4. Trainee returns to the main menu or starts training
Alternative Flow	**A1:** Instruction page fails to load → system displays an error message and reload option

Post-Condition	Trainee understands the training procedures and VR controls before starting the simulation.

Table 3.1.3. 3: Use Case Specification – View Instructions

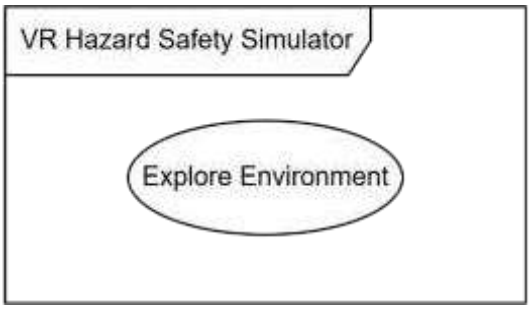


Figure 3.1.3.1.4: Use Case - Explore Environment

3.1.3.4 Use Case 4: Explore Environment

Use Case ID	UC-04
Use Case Name	Explore Environment
Description	Allows the trainee to explore the virtual warehouse environment before identifying hazards.
Actor	Trainee (New warehouse worker)
Pre-Conditions	I. Training session has started II. Virtual warehouse environment is loaded successfully
Basic Flow	1. Trainee enters the virtual warehouse 2. Uses VR controls to move around the environment 3. Observes warehouse areas and objects 4. Interacts with the environment to understand the surroundings

Alternative Flow	**A1:** Movement control does not respond → system displays a control error message **A2:** Environment fails to load completely → system reloads the warehouse scene
Post-Condition	Trainee successfully explores and becomes familiar with the virtual warehouse environment.

Table 3.1.3. 4: Use Case Specification –Explore Environment



Figure 3.1.3.1.5: Use Case - Identify Hazard

3.1.3.5 Use Case 5: Identify Hazard

Use Case ID	UC-05
Use Case Name	Identify Hazards
Description	Allows the trainee to identify hazards within the virtual warehouse environment during the training session.
Actor	Trainee (New warehouse worker)
Pre-Conditions	I. Training session is active II. Trainee is inside the virtual warehouse environment
Basic Flow	1. Trainee explores the warehouse environment 2. Observes objects and surrounding areas 3. Selects or points at identified hazards 4. System checks the selected hazard 5. System records the trainee's response and score 6. Training continues until all hazards are identified

Alternative Flow	**A1:** Trainee selects the wrong object → system displays incorrect feedback and deducts marks if applicable **A2:** Hazard interaction fails → system reloads the interaction function
Post-Condition	The identified hazards and trainee performance are successfully recorded by the system.

Table 3.1.3. 5: Use Case Specification – Identify Hazard

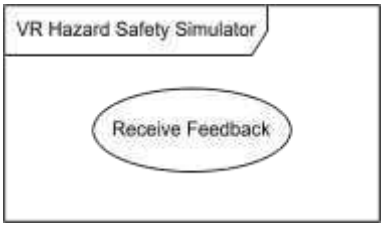


Figure 3.1.3.1.6: Use Case - Receive Feedback

3.1.3.6 Use Case 6: Receive Feedback

Use Case ID	UC-06
Use Case Name	Receive Feedback
Description	Allows the trainee to view training results, scores, and feedback after completing the hazard identification session.
Actor	Trainee (New warehouse worker)
Pre-Conditions	I. Training session has been completed II. System has recorded trainee performance
Basic Flow	1. Trainee completes the training session 2. System calculates the total score 3. System displays identified hazards and mistakes 4. Feedback and performance results are shown 5. Trainee reviews the results

Alternative Flow	**A1:** Score calculation error occurs → system recalculates the results automatically **A2:** Feedback page fails to load → system displays a reload option
Post-Condition	Training score and feedback are successfully displayed to the trainee.

Table 3.1.3. 6: Use Case Specification – Receive Feedback



Figure 3.1.3.1.7: Use Case - View Results

3.1.3.7 Use Case 7: View Result Summary

Use Case ID	UC-07
Use Case Name	View Result Summary
Description	Allows the trainee to view the final training result after completing the hazard identification session.
Actor	Trainee (New warehouse worker)
Pre-Conditions	III. Training session has been completed IV. System has calculated trainee performance
Basic Flow	1. Trainee completes the VR training session 2. System generates training results 3. System displays total score and identified hazards 4. Trainee reviews performance feedback and results
Alternative Flow	**A1:** Result data fails to load → system displays reload message **A2:** Score calculation error occurs → system recalculates the result automatically

Post-Condition	Training results and feedback are successfully displayed to the trainee.

Table 3.1.3. 7: Use Case Specification – View Result Summary

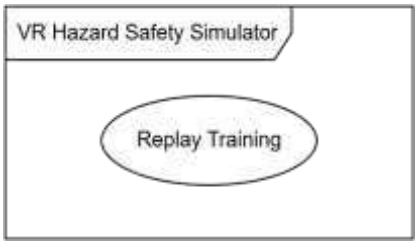


Figure 3.1.3.1.8: Use Case - Replay Training

3.1.3.8 Use Case 8: Replay Training

Use Case ID	UC-08
Use Case Name	Replay Training
Description	Allows the trainee to repeat the SecureSite VR training session for practice and improvement.
Actor	Trainee (New warehouse worker)
Pre-Conditions	Trainee has completed at least one training session VR simulator system is functioning properly
Basic Flow	1. Trainee selects “Replay Training” 2. System resets the previous training session 3. Virtual warehouse environment reloads 4. Training session starts again 5. Trainee repeats hazard identification activities
Alternative Flow	**A1:** System fails to reload training environment → system displays reload error message

	A2: Training data reset fails → system prompts trainee to restart the simulator
Post-Condition	A new VR training session is started successfully for the trainee.

Table 3.1.3. 8: Use Case Specification – Replay Training

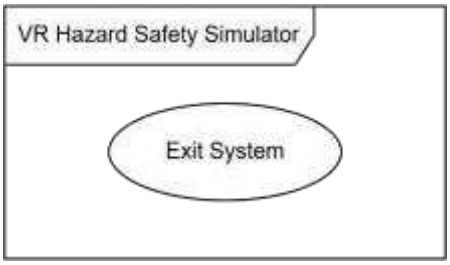


Figure 3.1.3.1.9: Use Case - Exit System

3.1.3.9 Use Case 9: Exit System

Use Case ID	UC-09
Use Case Name	Exit System
Description	Allows the trainee to close and exit the SecureSite VR safely.
Actor	Trainee (New warehouse worker)
Pre-Conditions	Trainee is using the VR simulator system
Basic Flow	1. Trainee selects “Exit System” 2. System asks for exit confirmation 3. Trainee confirms exit action 4. System closes the VR simulator application

Alternative Flow	**A1:** Trainee cancels exit action → system returns to the main menu **A2:** System fails to close properly → system displays error message and retries closing process
Post-Condition	The VR simulator system is successfully closed and the trainee exits the application.

Table 3.1.3. 9: Use Case Specification – Exit System

3.1.4 Sequence Diagram

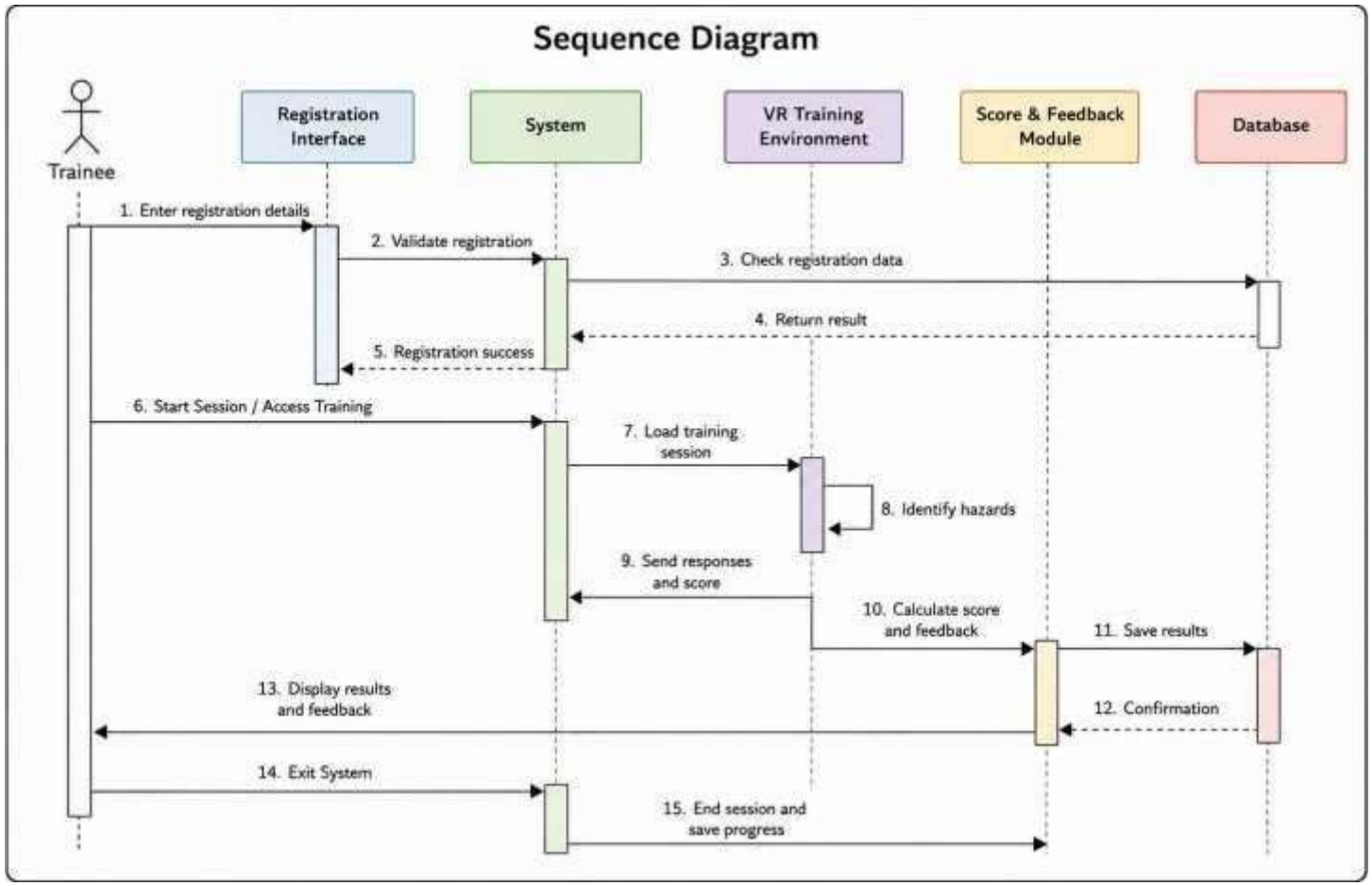


Figure 3.1.4: Sequence Diagram

3.1.5 Activity Diagram

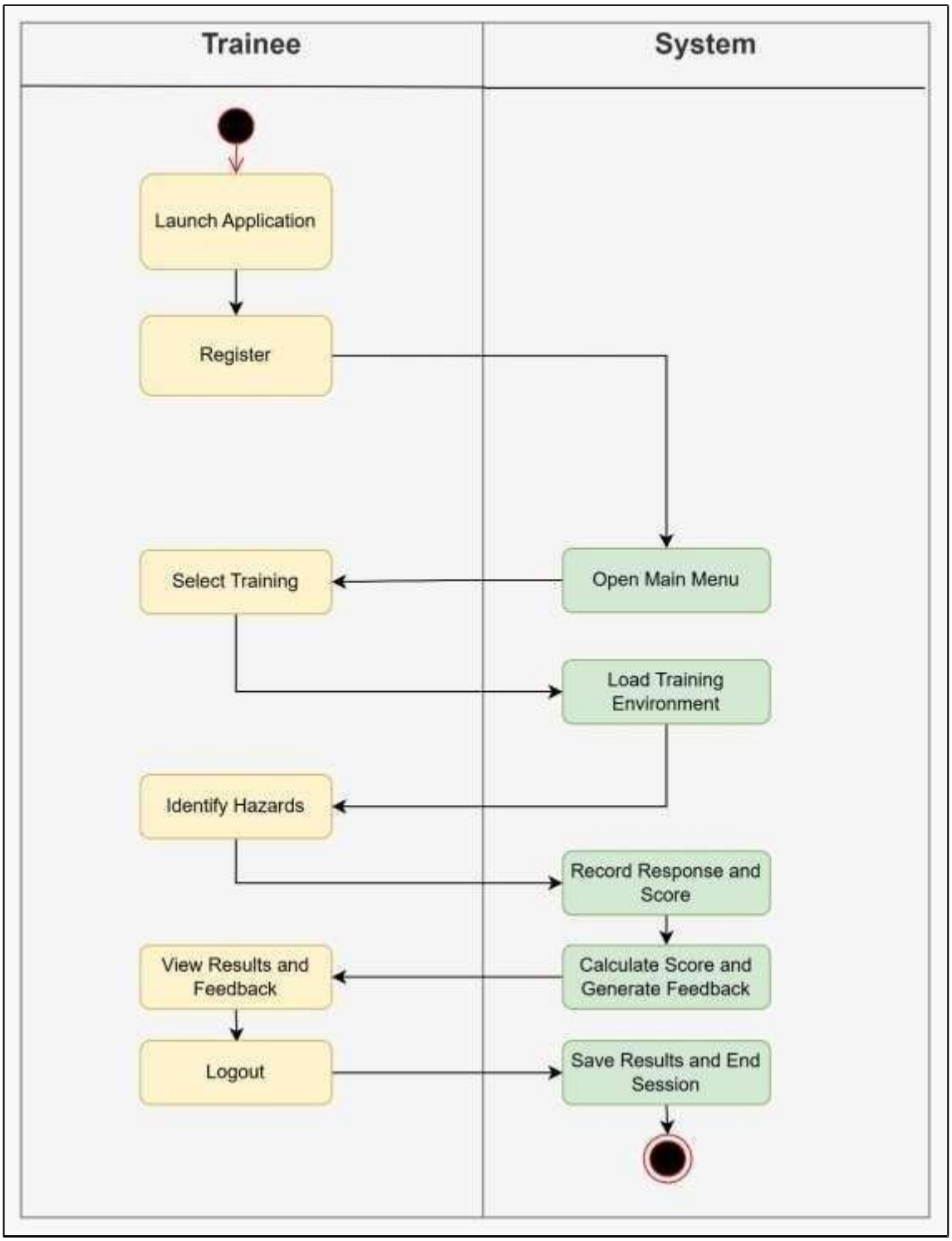


Figure 3.1.5: Activity Diagram

3.1.6 Class Diagram

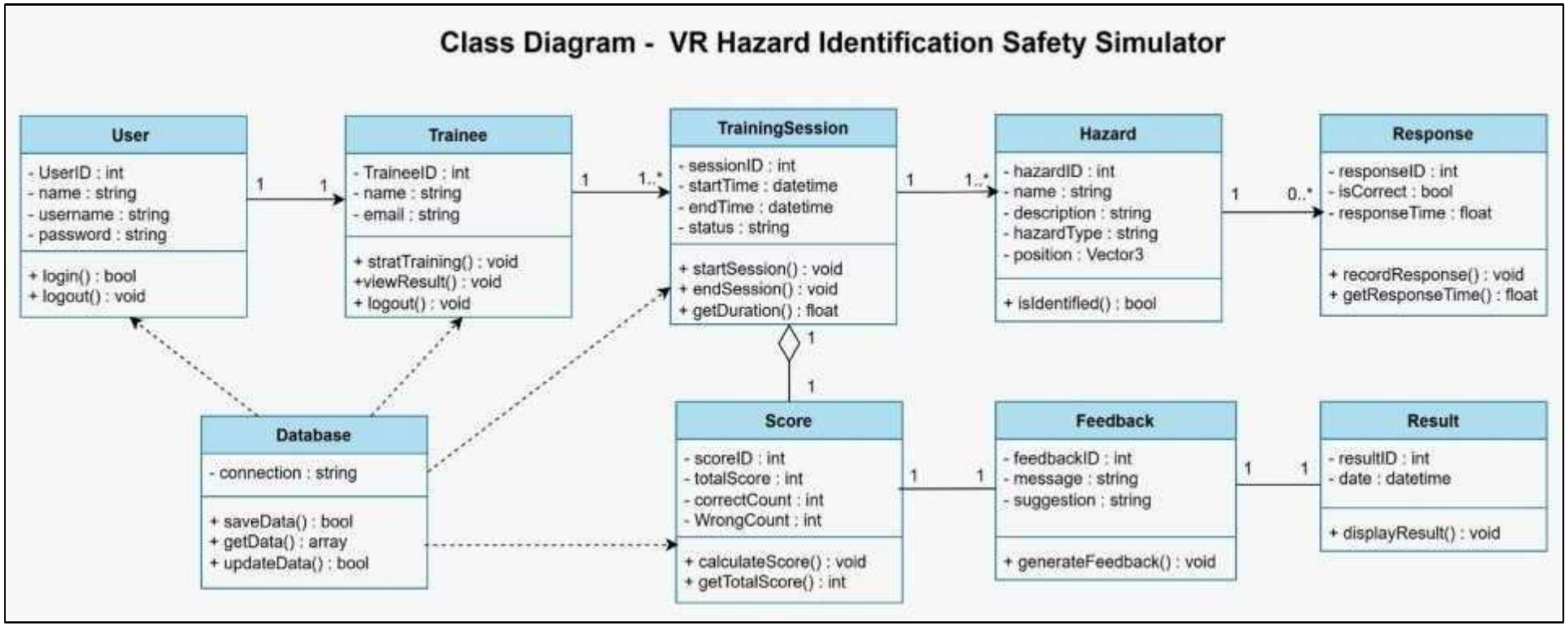


Figure 3.1.6: Class Diagram

3.1.7 FlowChart

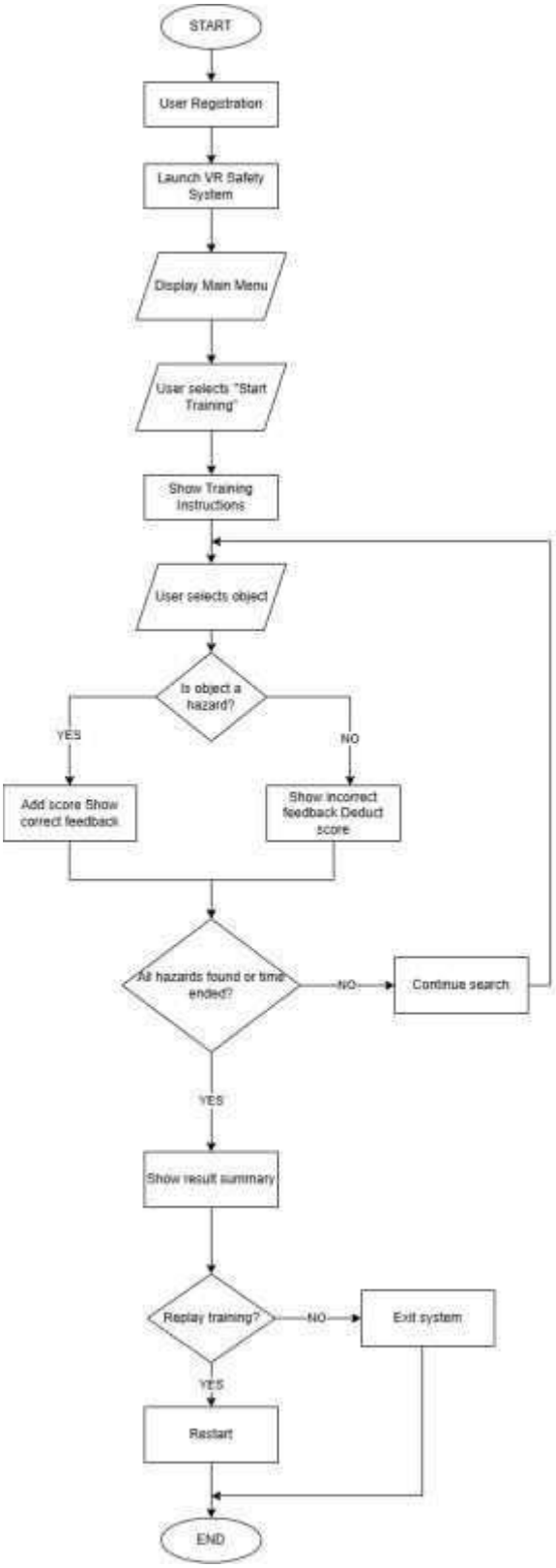


Figure 3.1.7: Flowchart Diagram

3.2 PROJECT SPECIFICATION

3.2.1 Software Use Specification

The software requirements for the development of the SecureSite VR are listed in Table below.

No.	Software	Version	Purpose
1	 Unity 3D	2022 or later	Main development platform for creating VR environment and simulation
2	 C#	-	Programming language used for scripting and system logic
3	 OpenXR	Latest	Integration between VR hardware and Unity

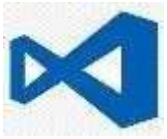
4	 Blender	3.0 or later	Creation and editing of 3D models and assets
5	 Visual Studio Code	Latest	Code development and debugging
6	 GitHub	Latest	Version control and project collaboration

Table 3.2.1: Software Requirements

Minimum Software Requirements

- Windows 10 or later
- Unity Hub installed
- Stable internet connection

3.2.2 Hardware Use Specification

The hardware requirements for running and developing the SecureSite VR are shown in Table below.

No.	Hardware	Image	Specifications
1	VR Headset		- Display: Fast-switch LCD - Resolution: 1832 × 1920 per eye - Refresh Rate: Up to 120Hz - Storage: 128GB - Tracking: Inside-out tracking - Controllers: Motion Tracking Sensors
2	Development Computer		- Processor: Intel i5 / Ryzen 5 or higher - RAM: Minimum 8GB (16GB recommended) - GPU: GTX 1060 or higher - Storage: SSD (256GB or more)

3	Monitor (Acer Nitro ED270R S3)		- Resolution: Full HD (1920×1080) - Size: 21” or above - 180Hz Refresh Rate - Curved VA Panel - 1500R curvature.
4	Mouse		- Standard USB/Wireless Input Devices
5	Keyboard		- Standard USB/Wireless Input Devices

Table 3.2.2: Hardware Requirements

3.3 PROJECT FEATURE

3.3.1 User Login

Users are required to log into the system before accessing the training simulation. Authentication can be implemented using a simple username and password system to track user progress and performance.

3.3.2 User Registration

New users must register by creating an account with basic details such as username and password. This allows the system to store user data and training records.

3.3.3 VR Environment Exploration

Users can explore a fully immersive virtual environment that simulates real-world workplaces such as construction or industrial sites. Movement and navigation are controlled using VR hardware.

3.3.4 Hazard Identification

Users are required to identify hazards within the environment, such as fire risks, exposed wires, or unsafe objects. This feature helps improve safety awareness.

3.3.5 Interactive Object Handling

Users can interact with objects using VR controllers, such as picking up tools or inspecting hazardous items within the environment.

3.3.6 Safety Training Module

The system provides guided instructions on proper safety procedures through text prompts, visual cues, or audio guidance to assist users during training.

3.3.7 Real-Time Feedback

The system provides immediate feedback based on user actions. Correct actions are acknowledged, while incorrect actions trigger warnings or hints.

3.3.8 Performance Evaluation

The system evaluates user performance based on:

- Number of hazards identified
- Time taken to complete the scenario
- Number of incorrect actions

3.3.9 Scenario-Based Training

Multiple training scenarios are provided to simulate different environments and hazard situations, allowing users to gain diverse learning experiences

3.4 PROJECT COSTING

The project costing outlines the estimated expenses required to develop SecureSite VR . The cost includes both hardware and software components. The use of free and open-source software helps to reduce the overall cost of development.

3.4.1 Required Items

No	Item	Description	Quantity	Unit Price (MYR)	Total Cost (MYR)
1	VR Headset (Meta Quest 2)	Used for immersive VR experience and testing Included with headset	1	1,500	1,500
2	Asset resources pack	3D modeling, animation, sound effect and system design	-	500	500
3	Monitor x2, Speaker, Mouse x2 and Keyboard x2.	Display, audio instructions, and controlling the simulator during the presentation.	7	1,960	1,960
Total Estimated Cost (RM)					3,960

Table 3.4.1: Required items

The total cost of the project is relatively low due to the use of free software tools. The main expenses are associated with hardware components such as the VR headset and development computer. The cost can be further reduced if existing hardware resources are utilized.

CHAPTER IV

TESTING METHODOLOGY

Testing methodology refers to the process used to evaluate whether the SecureSite VR functions correctly, meets project requirements, and provides a smooth learning experience for users. The testing process ensures that the system is stable, responsive, accurate, and user-friendly before deployment.

The project testing process focuses on system functionality, user interaction, hardware compatibility, and performance within the virtual reality environment.

4.1 TESTING MODEL

The testing model used for this project is based on the Agile Model. The Agile approach is suitable for developing the SecureSite VR system because the project involves continuous developments, testing and improvement of VR features and user interactions. The system is developed in small iterations, where each module is tested after implementation to ensure the functions work correctly and meet the project requirements.

The testing activities are conducted continuously throughout the development process to ensure all modules and features work correctly according to the system requirements.

Advantages of Using the Agile Model

i. Continuous testing and improvements during development
ii. Faster identification and fixing of system errors
iii. Flexible for adding or improving VR features
iv. Improves system quality and user experience
v. Encourages better collaborations and feedback throughout the project

4.1.1 Agile Model

The Agile Model is an iterative software development methodology that focuses on continuous development, testing, and improvement throughout the project lifecycle. This model is suitable for the SecureSite VR project because VR system development requires frequent testing, feedback, and modifications to improve user interaction, hazard simulation, and overall training effectiveness. Agile also allows the project to be developed in smaller and manageable phases called iterations or sprints.

The first phase is Requirement Analysis and Planning, where the project requirements, workplace safety objectives, and VR training goals are identified. Information is gathered through research and discussions to understand the problems faced in traditional safety training and determine the required system features and hazard scenarios.

The second phase is System Design, where the VR environment, user interface, hazard scenarios, and system workflow are designed. The warehouse environment, hazard placement, scoring system, and user interaction are planned carefully to ensure the simulator provides an effective learning experience.

Next is the Development and Implementation phase, where the SecureSite VR system is developed using suitable tools such as Unity and C# programming language. The virtual environment, hazard objects, interaction mechanisms, and training modules are created progressively during each iteration.

The Testing phase is conducted continuously throughout the development process to ensure the system functions correctly and meets the project requirements. Several testing methods such as functionality testing, usability testing, and performance testing are carried out to identify errors, improve system quality, and enhance user experience. Hazard detection accuracy, VR interaction, and scoring features are also tested regularly during development.

After completing each iteration, feedback and improvements are applied to enhance the system. This process allows modifications and additional features to be implemented more easily based on testing results and user evaluations.

Finally, the Deployment and Maintenance phase is carried out after the system is completed. The VR simulator is prepared for presentation, demonstration, and final use. Future improvements, bug fixes, and additional hazard scenarios can also be added to ensure the system remains effective and up to date.

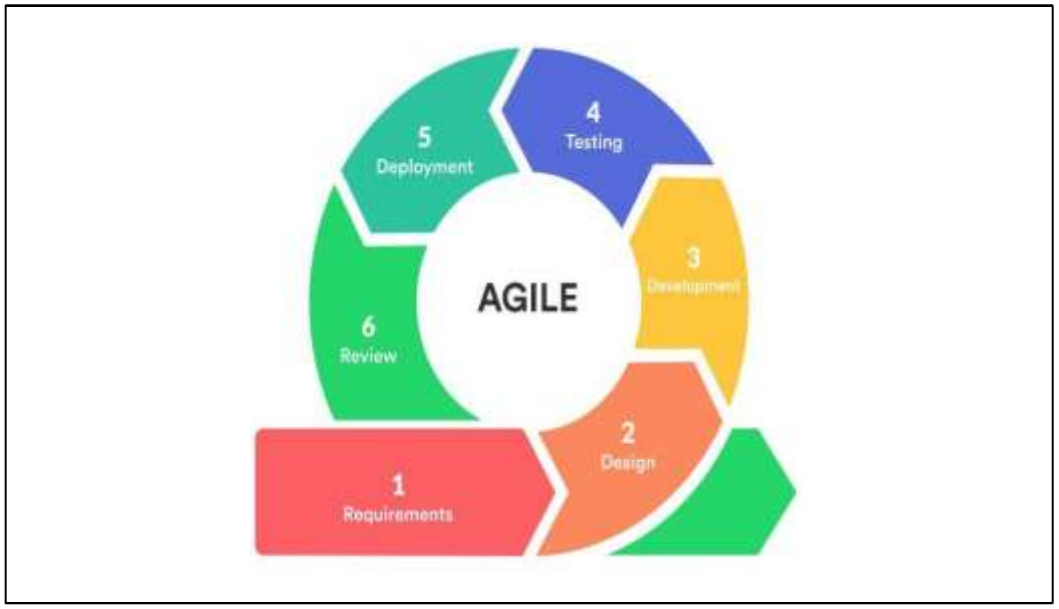


Figure 4.1.1: Agile Model

4.1.2 Testing Area

Several testing areas were identified to ensure the SecureSite VR operates effectively.

1. **Functional Testing**: Ensure buttons, menus, interactions, and VR features function correctly.

2. **User Interface Testing**: Verify that the interface is user-friendly and easy to navigate.

3. **VR Interaction Testing**: Test hand controls, object interaction, movement, and immersion within the VR environment.

4. **Performance Testing**: Ensure smooth frame rates and system stability during usage.

5. **Compatibility Testing**: Confirm compatibility between the VR headset, controllers, and development software.

6. **Safety Scenario Testing:** Verify that hazard simulations and safety procedures operate accurately.

7. **User Acceptance Testing**: Collect feedback from users to evaluate usability and effectiveness.

4.2 TESTING TOOLS

Testing tools are important to evaluate the functionality, performance, and reliability of the system. The tools used in this project include software development tools, debugging tools, and VR hardware testing equipment.

4.2.1 Software Testing Tools Used

Tool	Purpose
 Unity Game Engine	Used to develop, run, and test the VR environment and interactions.
 Visual Studio	Used for coding, debugging, and error checking.
 Meta Quest Link	Used to connect and test VR functionality with the headset.

 Oculus Debug Tool	Used to monitor VR performance and system stability.
 GitHub	Used for version control and collaborative project management.
 Google Forms	Used to collect user testing feedback and evaluation results.

Table 4.2.1: Software Testing Tools Used

4.2.2 Testing Type

The following testing types are going to be used throughout the project implementation process.

1. **Unit Testing**
   Unit testing is used to test individual modules, scripts, or components of the SecureSite VR to ensure each function works correctly on its own. This testing helps identify errors early in specific parts of the system before they are combined with other components.

2. **Integration Testing**
   Integration testing is used to check whether different parts of the VR system work together properly. It ensures that modules such as the VR environment, user interface, and hazard simulation components interact smoothly without any conflicts or data issues.

3. **System Testing**
   System testing is performed to evaluate the complete SecureSite VR as a whole. This testing ensures that all integrated components function correctly together and that the system meets the overall project requirements.

4. **Usability Testing**
   Usability testing is used to assess how easy and user-friendly the VR system is for users. It focuses on interface clarity, navigation, interaction within the VR environment, and overall user experience to ensure users can understand and use the system comfortably.

5. **Performance Testing**
   Performance testing is used to measure the responsiveness and stability of the VR system during operation. It evaluates frame rate performance, loading speed, and system smoothness to ensure the VR experience runs without lag or interruption.

6. **User Acceptance Testing (UAT)**

   User Acceptance Testing is conducted with actual users to confirm whether the SecureSite VR meets project objectives and user requirements. It ensures the system is suitable for real-world use and satisfies user expectations.

CHAPTER V

FINDING AND ANALYSIS

5.1 PROJECT MOCKUP

Project mockup refers to the proposed graphical user interface (GUI) design of SecureSite VR. The mockup is created to provide an early visualization of how the system interface, navigation flow, and interaction elements will appear before the full system implementation is completed.

The proposed mockup focuses on creating a simple, user-friendly, and interactive interface suitable for new industrial workers undergoing safety training. The GUI design emphasizes ease of navigation, hazard visibility, and clear feedback to improve user learning experience within the VR environment.

5.2 USER GUI SYSTEM PROPOSED

The User GUI System is designed for trainees who will use the VR simulator to perform hazard identification activities within the warehouse environment. The interface is developed with simple navigation and interactive elements to ensure users can easily understand and operate the system.

5.2.1 Proposed User GUI Components

1. Registration Interface



Figure 5.2.1.1: Registration Interface

The registration interface is displayed before users begin the VR training session. This interface allows users to enter basic information such as username and select the type of industrial environment they have worked in. The purpose of this interface is to simplify user identification and personalize the training experience based on the selected environment.

The interface includes:

i. Username input field
ii. Industrial environment dropdown selection
iii. Continue button

The design focuses on providing a simple and fast registration process to ensure users can access the training system easily without complicated setup procedures.

2. Main Menu Interface



Figure 5.2.1.2: Main Menu Interface

The main menu is the first navigation interface after registration. It allows users to:

i. Start training
ii. View instructions
iii. Access settings
iv. Exit the system

This interface provides simple navigation and helps users quickly access the required functions.

3. Instruction Interface

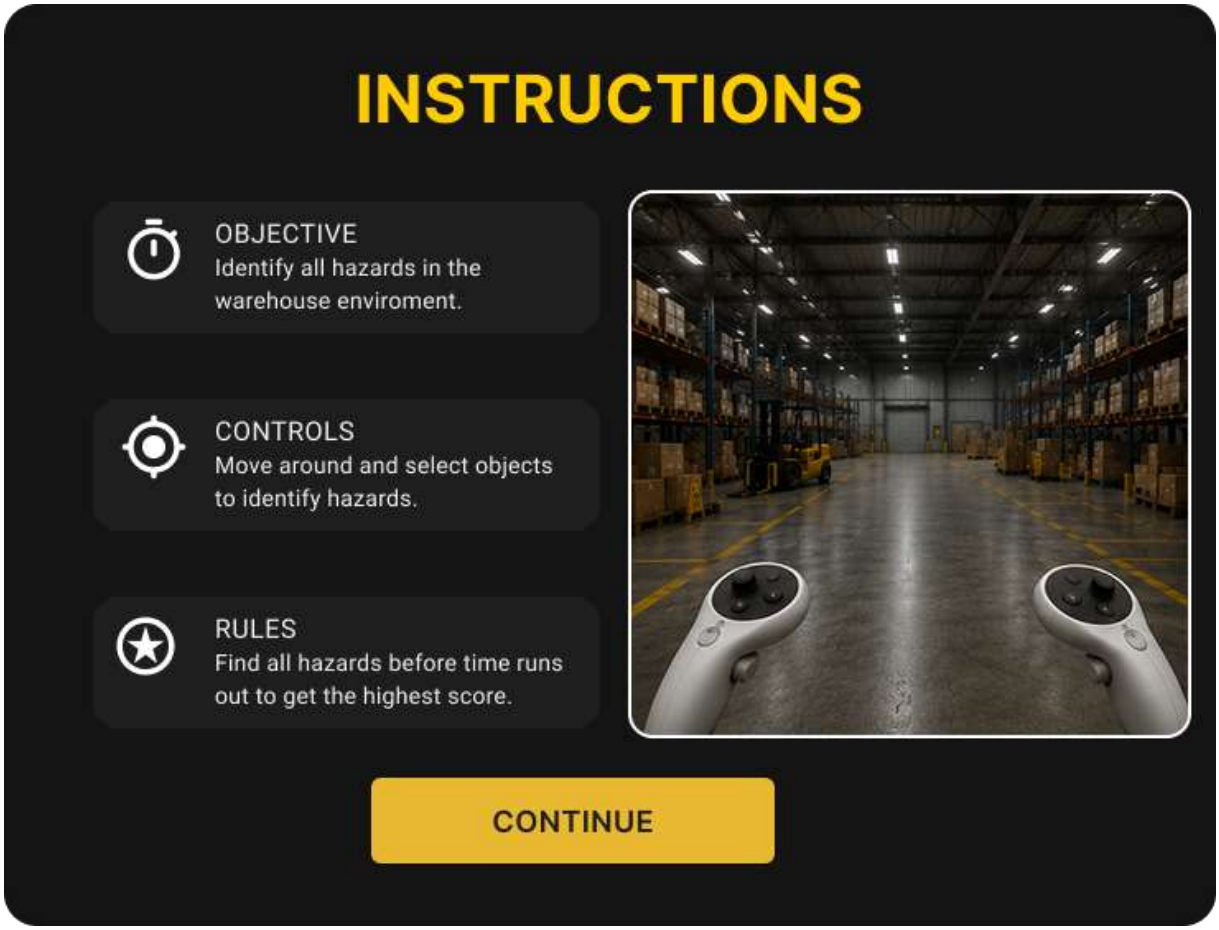


Figure 5.2.1.3: Instruction Interface

The instruction interface provides users with:

i. Training objectives
ii. VR controls and navigation guidance
iii. Hazard identification instructions

This helps users understand how to interact with the system before entering the simulation environment.

4. VR Training Environment Interface

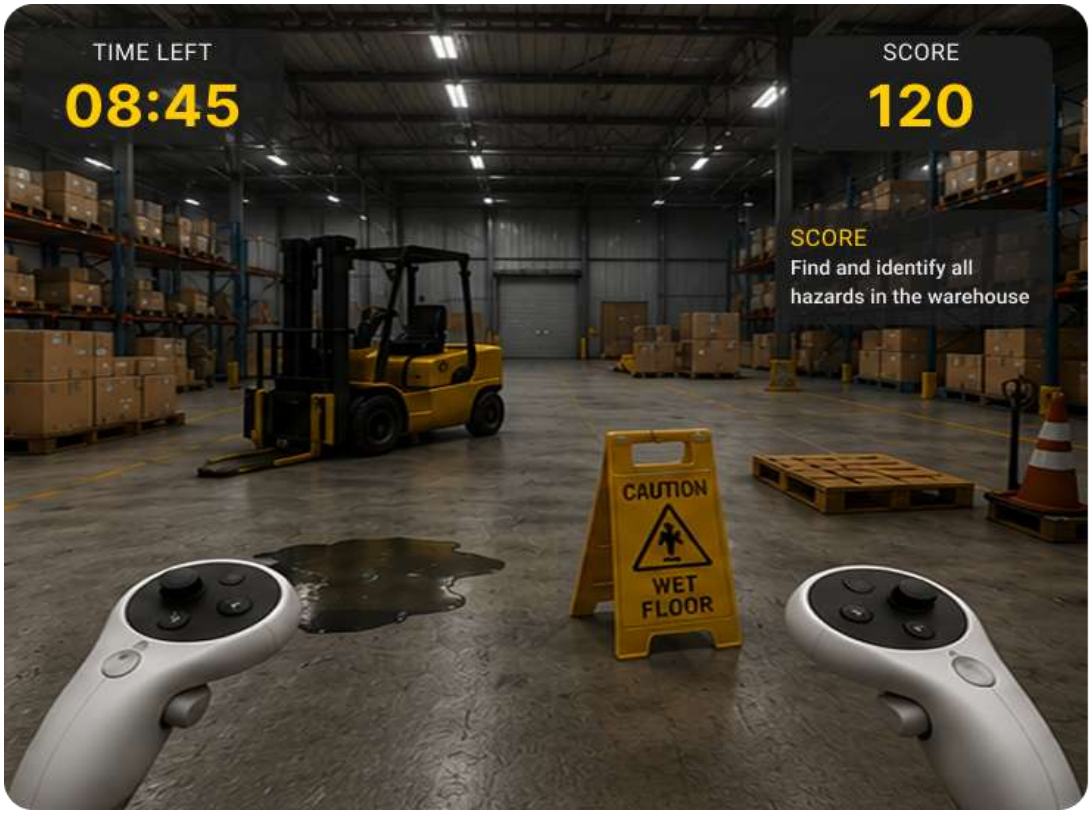


Figure 5.2.1.4:VR Training Environment Interface

The VR training environment interface displays the virtual warehouse where users perform hazard identification activities. Users can:

i. Explore surroundings
ii. Observe workplace conditions
iii. Interact with hazard objects

The interface also includes:

i. Timer display
ii. Current score
iii. Hazard objectives

This helps users monitor their training progress throughout the simulation.

5. Hazard Feedback Interface

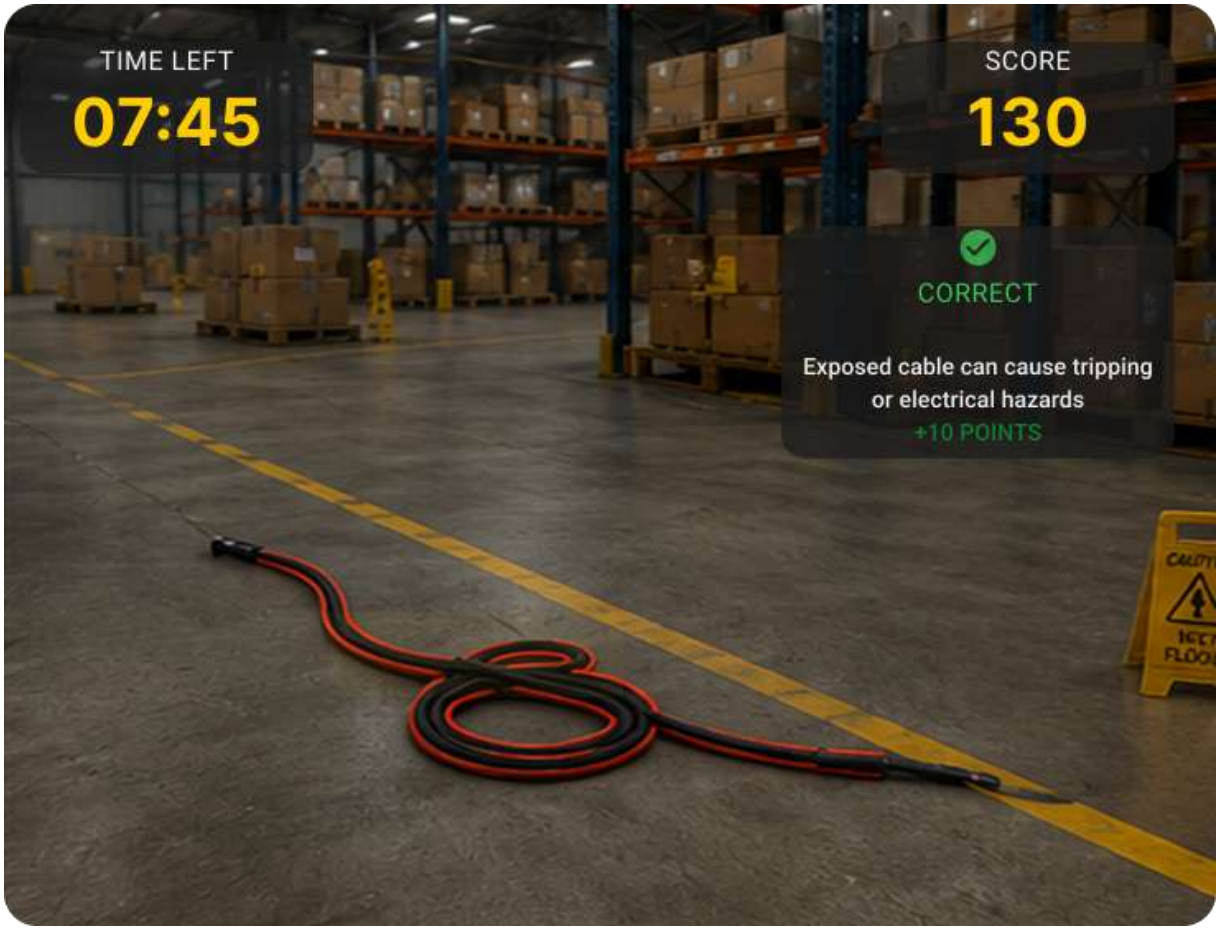


Figure 5.2.1.5: Hazard Feedback Interface

When a user selects an object, the system immediately displays feedback indicating whether the selected object is a hazard or not. Correct selections increase the score, while incorrect selections trigger warning feedback.

This interface helps improve user learning and hazard awareness through immediate response.

6. Result Summary Interface

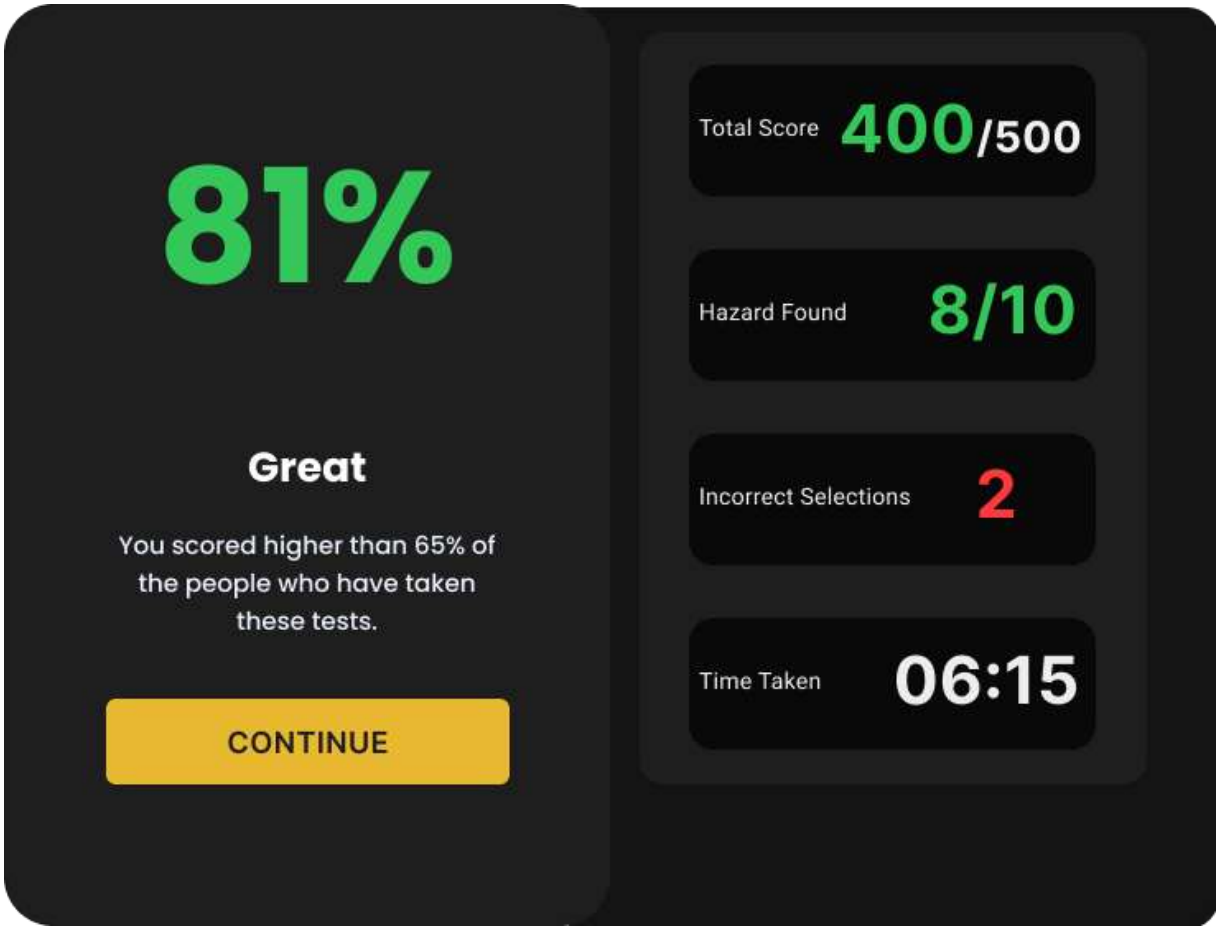


Figure 5.2.1.6: Result Summary Interface

After the training session is completed, the system displays a result summary containing:

i. Total score
ii. Number of hazards identified
iii. Incorrect selections
iv. Time taken

This interface allows users to evaluate their performance and identify areas for improvement.

7. Replay and Exit Interface

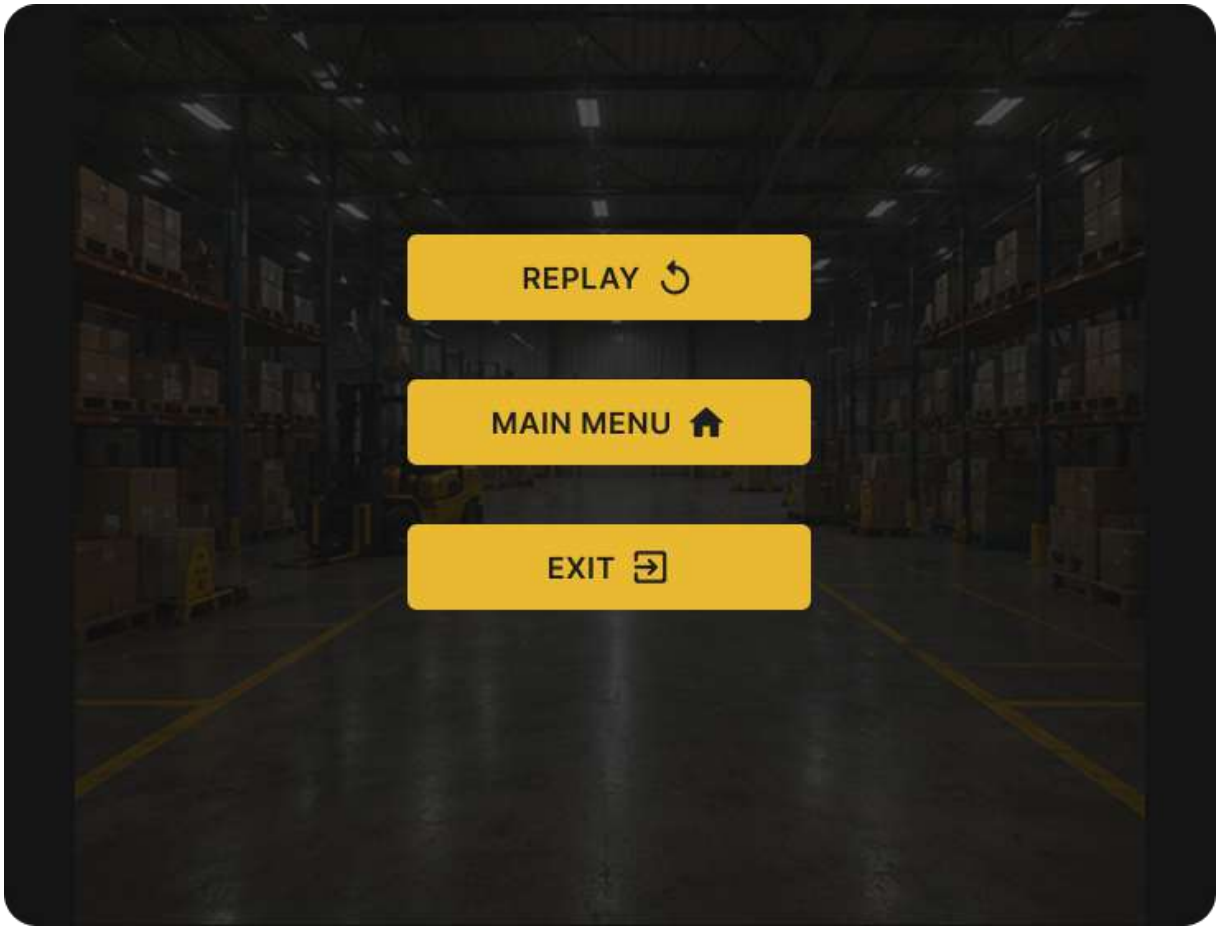


Figure 5.2.1.7: Replay and Exit Interface

Users can choose to:

i. Replay the training session
ii. Return to main menu
iii. Exit the system

This feature supports repeated learning and continuous practice for better hazard recognition skills.

5.3 ADMIN GUI SYSTEM PROPOSED

The Admin GUI System is designed to manage and monitor the training system. The administrator is responsible for maintaining hazard information, training settings, and user performance records.

5.3.1 Proposed Admin GUI Components

1. Admin Dashboard



Figure 5.3.1.1: Admin Dashboard

The dashboard provides an overview of:

i. Total training sessions
ii. User scores
iii. Hazard statistics
iv. System activity

This interface helps administrators monitor system usage and training performance.

2. Hazard Management Interface



Figure 5.3.1.2: Hazard Management Interface

Administrators can:

i. Add new hazards
ii. Edit hazard information
iii. Remove hazard objects
iv. Update hazard descriptions

This ensures the training content remains relevant and accurate.

3. User Performance Management

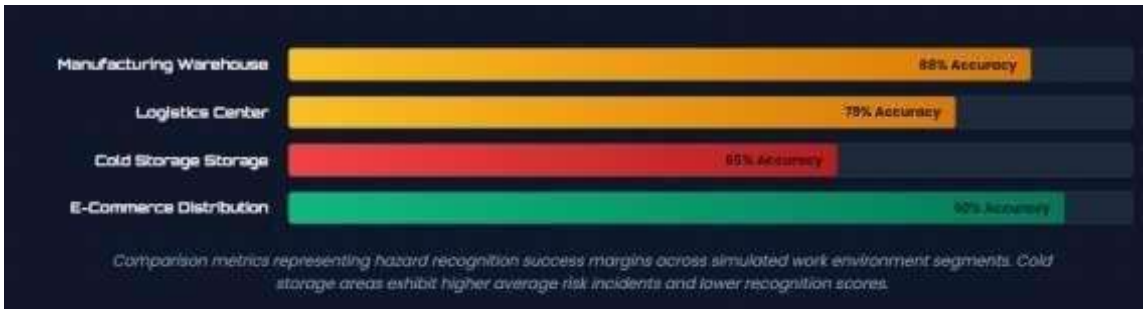


Figure 5.3.1.3: User Performance Management

The admin can view:

i. User training results
ii. Scores achieved
iii. Hazards missed
iv. Completion time

This helps evaluate user understanding and training effectiveness.

4. Training Settings Interface



Figure 5.3.1.4: Training Settings Interface

Administrators can configure:

i. Time limits
ii. Scoring rules
iii. Hazard quantity
iv. Difficulty settings

This feature allows the training session to be adjusted based on training requirements.

5. System Maintenance Interface



Figure 5.3.1.5: System Maintenance Interface

The maintenance interface allows administrators to:

i. Manage system updates
ii. Monitor application status
iii. Backup training data

This helps maintain system reliability and performance.

5.4 PROBLEMS ENCOUNTERED AND COUNTERMEASURES

During the development of the SecureSite VR, several technical problems were encountered. These issues mainly involved VR performance, hazard interaction accuracy, and data storage. The following sections explain the problems and the solutions implemented.

Problems Encountered	Countermeasures
The VR simulator experienced lag and unstable frame rates due to the large warehouse environment and many 3D objects. This caused discomfort during gameplay.	Static batching, baked lighting, and occlusion culling were implemented to optimize performance and improve frame rate stability.
Users had difficulty selecting smaller hazards such as exposed wires and floor spills using the VR controller pointer.	Larger colliders, a raycast snap system, and haptic feedback were added to improve hazard selection accuracy and interaction responsiveness.
The system initially failed to save user records and scores on the Meta Quest device because of storage restrictions.	Unity's persistent data path and JSON serialization were used to safely save and load user data locally on the headset.

Table 5.2: Problem Encountered and Countermeasures

5.5 RESULT AND DISCUSSION

The evaluation of the proposed SecureSite VR was conducted based on feedback collected from 41 respondents, consisting of students and individuals with some industrial exposure. The purpose of this evaluation was to understand user perceptions of the usefulness, usability, and training potential of a VR-based safety training system prior to full development.

Overall, the feedback indicated strong support for the idea of implementing VR technology in workplace safety training. Respondents generally agreed that a VR-based approach would provide a more engaging and interactive learning experience compared to traditional methods such as lectures or static training materials. The immersive nature of VR was highlighted as a key advantage, as it allows users to actively explore a simulated environment and identify hazards in a more realistic context.
In terms of usability expectations, participants emphasised the importance of having a clear and simple interface within the VR environment. It was generally expected that interactions such as selecting or identifying hazards should feel intuitive and responsive, provided that appropriate design and optimisation techniques are applied. Comfort during VR usage was also considered an important factor, with respondents noting that performance stability and well-designed movement mechanics would be essential in reducing discomfort such as motion sickness.

Furthermore, participants expressed that a VR training system could be particularly beneficial for improving hazard recognition skills. The ability to simulate real-world industrial environments in a safe and controlled manner was seen as a major advantage, especially for new or inexperienced workers. This would allow users to learn through experience rather than passive observation, which was perceived to enhance understanding and retention of safety knowledge.

The findings also suggest that VR-based training has strong relevance in situations where real-life exposure to hazards is either impractical, costly, or unsafe. By providing a controlled virtual environment, users can repeatedly practise identifying hazards without any physical risk. Overall, the results indicate that there is a clear interest in and positive perception of VR as a modern tool for workplace safety education, supporting the direction of this project.

CHAPTER VI

CONCLUSION

6.1 CONCLUSION

This project proposes the development of a SecureSite VR aimed at improving workplace safety training for new industrial workers. The system is designed to provide a more immersive and interactive learning experience compared to traditional safety training methods, allowing users to explore a virtual industrial environment and identify common workplace hazards in a controlled setting.

The proposed simulator focuses on enhancing hazard awareness by placing users in realistic scenarios where they can actively observe and respond to potential safety risks. Immediate feedback and performance evaluation features are intended to support learning by reinforcing correct hazard identification behaviour during training sessions. The development approach for the system follows a structured methodology using the Agile model, with implementation planned using Unity, OpenXR, and VR hardware such as Meta Quest devices to ensure compatibility with modern immersive technologies.

Overall, findings from user feedback suggest that there is strong interest and positive perception toward VR-based safety training systems. The proposed solution is expected to improve engagement, provide a safer learning environment, and enhance the effectiveness of workplace safety education.

While the system is still in its proposal stage, it demonstrates clear potential as an alternative training approach. Future development may expand the system into multiple environments and improve scalability, interactivity, and data management capabilities to better support industrial training needs.

6.2 RECOMMENDATIONS

1. **Expansion of Training Environments and Scenarios**

   One recommendation for future improvement is to expand the variety of training environments and hazard scenarios within the VR system. Currently, workplace safety training is often limited to a single simulated setting, which may restrict the depth of user experience. By introducing additional environments such as construction sites, manufacturing plants, and chemical handling areas, users would be exposed to a wider range of realistic workplace conditions. This would enhance the training experience by improving adaptability and preparing users for different types of industrial hazards.

2. **Cloud-Based Data Storage and User Management**

   Another improvement is the integration of cloud-based storage and an online database system for managing user data. This would allow training records, performance history, and user progress to be stored more efficiently and accessed remotely. In addition, it would enable administrators or trainers to monitor multiple users more effectively, improving scalability and making the system more suitable for institutional or organizational deployment.

3. **Enhanced User Experience and Adaptive Learning Features**

   A further recommendation is to improve the overall user experience by incorporating advanced features such as adaptive difficulty levels, multilingual support, and detailed performance analytics. Adaptive difficulty would allow the system to adjust based on user performance, ensuring that training remains challenging but not overwhelming. Multilingual support would improve accessibility for a wider user base, while detailed feedback reports would help users better understand their strengths and areas for improvement, ultimately enhancing the learning effectiveness of the system.

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APPENDIXES

APPENDIX A

FYP01 GANTT CHART

	Task	Start Date	End Date	Duration (days)
2	Individual Proposal	1/19/2026	2/5/2026	17
3	Select Group Project Title	2/9/2026	2/18/2026	9
4	Pre-Approval Presentation	2/23/2026	3/5/2026	10
5	Feasibility Study (Chapter 1 & 2)	3/10/2026	3/14/2026	4
6	Project Design & Development (Chapter 3)	3/10/2026	4/3/2026	24
7	Progress Presentation	4/6/2026	4/16/2026	10
8	Proposal Report	4/20/2026	4/30/2026	10
9	Testing Methodology (Chapter 4)	5/1/2026	5/6/2026	6
10	Conclusion and Result (Chapter 5 & 6)	5/8/2026	5/13/2026	5
11	Final Report Submission	5/22/2026	5/23/2026	1
12	Submission of Marks	6/5/2026	6/6/2026	1
13	Final Marks Entry	6/19/2026	6/20/2026	1

